A PROBABILITY MEASURE ON HOMOTOPY & HOMOLOGY CLASSES VIA BROWNIAN LOOPS

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1. Introduction

2. Preliminaries

2.1. **The Brownian loop measure.** Let (X, g) be a complete orientable Riemannian surface, with or without boundary. When $\partial X \neq \emptyset$ we impose Dirichlet boundary conditions, so that Brownian motion is killed on first hitting the boundary. Brownian motion on X (run at speed 2) is the diffusion process generated by $-\Delta_X$, where

$$\Delta_X = -\mathrm{div}_g \, \mathrm{grad}_g$$

is the positive Laplace–Beltrami operator. We adopt this sign convention so that the spectrum of Δ_X lies in $[0, \infty)$. For $t > 0$ and $x, y \in X$, we write $p_X(t, x, y)$ for the heat kernel (i.e., transition density of Brownian motion), given as the integral kernel of the heat semigroup $e^{-t\Delta_X}$ via

$$(e^{-t\Delta_X} f)(x) = \int_X p_X(t, x, y)\, f(y)\; d\mathrm{vol}_g(y).$$

When obvious, we will drop the subscript X. For $x \in X$ and $t > 0$, we denote by $\mathbb{W}_x^t$ the law of Brownian motion started at x on the space of continuous paths $C([0, t], X)$, and by $\mathbb{W}_{x\to y}^t$ the unnormalised bridge measure from x to y in time t, defined by disintegrating $\mathbb{W}_x^t$ with respect to the endpoint

$$\mathbb{W}_x^t = \int_X \mathbb{W}_{x\to y}^t \; d\mathrm{vol}_g(y).$$

The bridge measure has total mass $|\mathbb{W}_{x\to y}^t| = p(t, x, y)$ and $\mathbb{W}_{x\to y}^t / p(t, x, y)$ is the usual conditional law given $\omega(t) = y$.

Let $\mathcal{C}_X^*$ denote the space of parametrised, oriented, rooted loops

$$\mathcal{C}_X^* := \big\{(t, \omega) : t \geq 0,\ \omega \in C([0, t]; X),\ \omega(0) = \omega(t)\big\}.$$

The slice $\{t = 0\}$ consists of constant point-loops and rooted means we distinguish a basepoint of the loop.

Definition 2.1 (Brownian loop measure)**.** The rooted Brownian loop measure on X is the infinite but σ-finite measure

$$\mu_X^* := \int_0^\infty \frac{dt}{t} \int_X \mathbb{W}_{x\to x}^t \; d\mathrm{vol}_g(x).$$

The weight dt/t is the multiplicative Haar measure on $(0, \infty)$ and provides a scale-invariant way to aggregate loops over all durations. By the short-time on-diagonal heat kernel asymptotic, $p(t, x, x) \sim 1/(4\pi t)$ as $t \downarrow 0$, the integrand diverges like $1/t^2$ near zero.

To pass from rooted parametrised loops to unrooted unparametrised ones we identify (t, ω) and $(s, \tilde{\omega})$ in $\mathcal{C}_X^*$, written $(t, \omega) \sim (s, \tilde{\omega})$, whenever there is an increasing continuous bijection

$i\colon [0,t]/0{\sim}t \to [0,s]/0{\sim}s$ of the parametrising circles such that $\tilde{\omega}(i(r)) = \omega(r)$ for all r. This forgets both the root and the time parametrisation while remembering the orientation. Such maps are generated by the circular time-shift action, where for $t > 0$

$$\mathrm{shift}_{s_0}(t,\omega) := (t, \omega_{s_0}), \qquad \omega_{s_0}(s) := \omega(s + s_0 \pmod t).$$

The Brownian loop measure μ_X is the pushforward of μ_X^* to the quotient $\mathcal{C}_X := \mathcal{C}_X^*/\sim$, the space of unparametrised, oriented, unrooted loops.

The Brownian loop measure has two fundamental properties (see [LW04, Section 4]). The first, restriction, says that if $X' \subset X$ is open, then $\mu_{X'}$ is the restriction of μ_X to loops contained in X'. More precisely, if $X' \subset X$, then $d\mu_{X'}(\eta) = \mathbb{1}_{\eta \subset X'} d\mu_X(\eta)$. The second, conformal invariance, says that replacing g by a conformally equivalent metric $e^{2\sigma}g$ leaves μ_X unchanged. Consequently μ_X depends only on the conformal class $[g]$ and we can take X to be a Riemann surface.

This conformal invariance property is particular to Brownian motion in 2D; for a subordinate process, the associated operator depends on the metric g itself, so we work with (X, g) as a Riemannian surface.

2.2. **Dirichlet form loop measures.** We now develop loop measures for processes associated with regular symmetric Dirichlet forms, following Le Jan's approach in the discrete case [LJ11].

Let (E, m) be a locally compact separable metric space equipped with a Radon measure m of full support, write $\langle\cdot,\cdot\rangle$ for the inner product of $L^2(E, m)$, and let $(\mathcal{E}, \mathcal{F})$ be a regular symmetric Dirichlet form on $L^2(E, m)$. Thus $\mathcal{E}$ is a non-negative definite symmetric bilinear form on a dense linear subset $\mathcal{F}$ of $L^2(E, m)$, closed and Markovian. See [FOT11] for a detailed explanation of the above, and note that the non-symmetric theory can be found in [MR92], but has fewer connections to Markov processes.

Such a form determines a non-negative definite self-adjoint operator A on $L^2(E, m)$, with $\mathcal{F} = \mathrm{Dom}(A^{1/2})$ and $\mathcal{E}(u, v) = \langle A^{1/2}u, A^{1/2}v\rangle$, so that

$$\mathcal{E}(u, v) = \langle Au, v\rangle, \qquad u \in \mathrm{Dom}(A),\ v \in \mathcal{F},$$

and hence a strongly continuous semigroup e^{-tA}. By the classic correspondence between regular symmetric Dirichlet forms and Markov (more precisely Hunt) processes [Fuk71], $(\mathcal{E}, \mathcal{F})$ determines such a process on E, unique up to quasi-everywhere equivalence. We assume throughout that the semigroup admits a jointly measurable symmetric transition density $p^{\mathcal{E}}(t, x, y)$ with respect to m,

$$(e^{-tA}f)(x) = \int_E p^{\mathcal{E}}(t, x, y)\, f(y)\, m(\,dy).$$

2.2.1. *The Dirichlet form loop measure.* From now on we fix $E = X$, a Riemannian surface, and let $(\mathcal{E}, \mathcal{F})$ denote a regular symmetric Dirichlet form on $L^2(X, \mathrm{vol}_g)$ with transition density $p^{\mathcal{E}}(t, x, y)$.

The construction of Definition 2.1 carries over to the Dirichlet form setting, with continuous paths replaced by càdlàg paths to accommodate jumps. For $t > 0$ and $x \in X$, disintegration gives $\mathbb{W}_x^{t,\mathcal{E}} = \int_X \mathbb{W}_{x\to y}^{t,\mathcal{E}}\, d\mathrm{vol}_g(y)$, where the unnormalised bridge measure $\mathbb{W}_{x\to y}^{t,\mathcal{E}}$ on the space of càdlàg paths $D([0,t]; X)$ has mass $p^{\mathcal{E}}(t, x, y)$. The space of parametrised, oriented, rooted loops is

$$\mathcal{C}_X^* := \{(t, \omega) : t \geq 0,\ \omega \in D([0,t]; X),\ \omega(0) = \omega(t)\},$$

and the quotient $\mathcal{C}_X := \mathcal{C}_X^*/\sim$ is the space of unparametrised, oriented, unrooted loops, where $\sim$ is the equivalence of Section 2.1, generated by circular time-shifts together with increasing continuous reparametrisations, both of which preserve the càdlàg property. From now on $\mathcal{C}_X^*$ and $\mathcal{C}_X$ denote càdlàg loop spaces, which contain their continuous counterparts.

Definition 2.2 (Dirichlet form loop measure)**.** The parametrised loop measure on $\mathcal{C}_X^*$ is the σ-finite measure

$$\mu_X^{*,\mathcal{E}} := \int_0^\infty \frac{dt}{t} \int_X \mathbb{W}_{x\to x}^{t,\mathcal{E}} \, d\mathrm{vol}_g(x).$$

It is invariant under the circular time-shift ([LJ11, Chapter 2]), and the loop measure $\mu_X^{\mathcal{E}}$ is its pushforward to $\mathcal{C}_X$.

The mass of $\mu_X^{*,\mathcal{E}}$ is $\int_0^\infty \frac{1}{t} \int_X p^{\mathcal{E}}(t,x,x)\, d\mathrm{vol}_g(x)\, dt$. Moreover, for an open set $X' \subset X$, the part form $\mathcal{E}_{X'}$ is associated with the process killed on leaving X', so its bridge measures are the ambient ones restricted to paths that stay in X'. The restriction property therefore holds.

2.3. **Bernstein functions and subordination.** We now show a way to produce new Dirichlet forms from a given one via subordination. Our main references are [SSV12] and [FOT11].

2.3.1. *Bernstein functions and subordinators.* A function $\phi\colon (0,\infty) \to [0,\infty)$ is a Bernstein function if it is C^∞ and $(-1)^{n-1}\phi^{(n)}(\lambda) \geq 0$ for all $n \geq 1$ and $\lambda > 0$. Every Bernstein function admits a unique Lévy–Khintchine representation

$$\phi(\lambda) = a + b\lambda + \int_0^\infty (1 - e^{-\lambda s})\, \nu(\, ds), \qquad \lambda > 0, \tag{1}$$

where $a \geq 0$ is the killing rate, $b \geq 0$ the drift, and ν the Lévy measure on $(0,\infty)$ satisfying $\int_0^\infty (1 \wedge s)\, \nu(\, ds) < \infty$.

A subordinator S_t is an increasing Lévy process on $[0,\infty)$, possibly killed at a constant rate, whose Laplace exponent is a Bernstein function ϕ,

$$\mathbb{E}[e^{-\lambda S_t}] = e^{-t\phi(\lambda)}, \qquad \lambda > 0,\ t \geq 0. \tag{2}$$

We write ψ_t^ϕ for the law of S_t on $[0,\infty)$. When $a = 0$ the subordinator is conservative and ψ_t^ϕ is a probability measure; when $a > 0$ it is killed at rate a and $|\psi_t^\phi| = e^{-at}$.

Assumption 2.3. Throughout we assume $b > 0$ or $\nu(0,\infty) = \infty$; equivalently, $S_t > 0$ almost surely for every $t > 0$, so that $\psi_t^\phi(\{0\}) = 0$. This excludes compound Poisson subordinators, whose semigroups don't admit a transition density.

Remark 2.4. Subordination and time change by a positive continuous additive functional (PCAF) both produce a new Markov process by running the original one on a different clock, but they act differently on the underlying data $(\mathcal{E}, \mathcal{F}, \mathrm{vol}_g)$. A subordinator S_t is independent of the original process and not adapted to its filtration; it replaces the generator $-A$ by $-\phi(A)$, which changes the associated Dirichlet form while keeping the reference measure vol_g. If the Lévy measure of ϕ is non-trivial, the subordinate process has jumps even when the original process is a diffusion. By contrast a PCAF clock is built from the process's own trajectory and is adapted to its filtration; the resulting time change reparametrises the same path, so continuity of paths is preserved, but the reference measure is replaced by the Revuz measure of the additive functional. See [RY99, Chapter 10]. A canonical example of the latter is Liouville Brownian motion [GRV16], where planar Brownian motion B_t is time-changed by the inverse of the "quantum clock" $F_t = \int_0^t e^{\gamma\varphi(B_s)}\, ds$ (with the exponential of the Gaussian free field φ on X

defined via Gaussian multiplicative chaos), giving $Z_t = B_{F^{-1}(t)}$, where $\gamma \in [0, 2)$ is the coupling constant.

Example 2.5 (Important Bernstein functions)**.** The following play a role in this paper.

Resulting process	$\phi(\lambda)$	(a, b, ν)	**Law** ψ_t^ϕ
Brownian motion	λ	$(0, 1, 0)$	δ_t
Brownian with killing ($\kappa > 0$)	$\lambda + \kappa$	$(\kappa, 1, 0)$	$e^{-\kappa t}\,\delta_t$
α-stable ($\alpha \in (0, 2)$)	$\lambda^{\alpha/2}$	$(0, 0, \nu_\alpha)$	density $\eta_t^\alpha(s) = t^{-2/\alpha}\, g_{\alpha/2}(s\, t^{-2/\alpha})$

Here $\nu_\alpha(\,ds) = \frac{\alpha/2}{\Gamma(1-\alpha/2)}\, s^{-1-\alpha/2}\,ds$, and $g_{\alpha/2}$ is the density of the standard $\alpha/2$-stable distribution on $(0, \infty)$. See [BBK+09, Chapter 5] for the explicit details. The framework also applies to other processes, such as gamma, inverse Gaussian, and relativistic stable subordinators.

2.3.2. *Subordination of semigroups and Dirichlet forms.* Let $(\mathcal{E}, \mathcal{F})$ be a regular symmetric Dirichlet form on $L^2(X, \mathrm{vol}_g)$. Given a Bernstein function ϕ with associated subordinator independent of the underlying process, the subordinate semigroup is

$$e^{-t\phi(A)} = \int_{[0,\infty)} e^{-sA}\, \psi_t^\phi(\,ds), \qquad t \geq 0, \tag{3}$$

where ψ_t^ϕ is the law of the subordinator at time t. Its operator is

$$\phi(A) = aI + bA + \int_0^\infty (I - e^{-sA})\, \nu(\,ds),$$

and hence the generator of the subordinate process is $-\phi(A)$. The subordinate process has transition density

$$p^\phi(t, x, y) = \int_{[0,\infty)} p^{\mathcal{E}}(s, x, y)\, \psi_t^\phi(\,ds). \tag{4}$$

Thus the subordinate kernel is obtained by averaging the original heat kernel against the law of the independent subordinator.

The subordinate process is again a vol_g-symmetric Hunt process associated with a regular symmetric Dirichlet form $(\mathcal{E}^\phi, \mathcal{F}^\phi)$ on $L^2(X, \mathrm{vol}_g)$, given by

$$\mathcal{E}^\phi(u, u) = a\, \|u\|^2 + b\, \mathcal{E}(u, u) + \int_0^\infty \big(\|u\|^2 - \langle e^{-sA}u, u\rangle\big)\, \nu(\,ds), \tag{5}$$

where $\|\cdot\|$ and $\langle\cdot,\cdot\rangle$ denote the $L^2(X, \mathrm{vol}_g)$ norm and inner product, and the domain is

$$\mathcal{F}^\phi := \{u \in L^2(X, \mathrm{vol}_g) : \mathcal{E}^\phi(u, u) < \infty\}.$$

2.3.3. *Application to the hyperbolic plane.* We apply the above theory to Brownian motion on the hyperbolic plane $\mathbb{H}^2$, whose Riemannian volume measure vol_g is the hyperbolic area measure, written ρ. Given a Bernstein function ϕ, the subordinate Dirichlet form $(\mathcal{E}^\phi, \mathcal{F}^\phi)$ on $L^2(\mathbb{H}^2, \rho)$ has operator $\phi(\Delta_{\mathbb{H}^2})$ and heat kernel

$$p_{\mathbb{H}^2}^\phi(t, z, w) = \int_{[0,\infty)} p_{\mathbb{H}^2}(s, z, w)\, \psi_t^\phi(\,ds), \qquad z, w \in \mathbb{H}^2. \tag{6}$$

Here $p_{\mathbb{H}^2}(s, z, w)$ is the Brownian heat kernel. Since $\Delta_{\mathbb{H}^2}$ is $\mathrm{PSL}(2, \mathbb{R})$-invariant, so is $p_{\mathbb{H}^2}^\phi(t, z, w)$.

Example 2.6. For $\phi(\lambda) = \lambda + \kappa$ with $\kappa > 0$, the subordinate form is

$$\mathcal{E}^{\kappa}(f, f) = \int_{\mathbb{H}^2} |\nabla f|^2 \, d\rho + \kappa \int_{\mathbb{H}^2} f^2 \, d\rho,$$

and the transition density is

$$p^{\kappa}_{\mathbb{H}^2}(t, z, w) = e^{-\kappa t} \, p_{\mathbb{H}^2}(t, z, w).$$

Example 2.7. For $\phi(\lambda) = \lambda^{\alpha/2}$ with $\alpha \in (0, 2)$, the operator is the fractional Laplacian $\Delta^{\alpha/2}_{\mathbb{H}^2}$ and the heat kernel is

$$p^{\alpha}_{\mathbb{H}^2}(t, z, w) = \int_0^\infty p_{\mathbb{H}^2}(s, z, w) \, \eta^{\alpha}_t(s) \, ds,$$

where η^{α}_t is the density of the $\alpha/2$-stable subordinator law ψ^{ϕ}_t. The process is pure-jump for all $\alpha \in (0, 2)$ and the boundary case $\alpha = 2$ recovers Brownian motion.

2.4. **Subordinate Brownian loop measure.** Fix a Bernstein function ϕ, and let $(\mathcal{E}^{\phi}, \mathcal{F}^{\phi})$ denote the subordinate Dirichlet form on $L^2(X, \mathrm{vol}_g)$ from Section 2.3, with semigroup $e^{-t\phi(A)}$ and transition density $p^{\phi}(t, x, y)$. The associated bridge measures have total mass $|\mathbb{W}^{t,\phi}_{x \to y}| = p^{\phi}(t, x, y)$.

Definition 2.8 (Subordinate Brownian loop measure)**.** Applying Definition 2.2 to $(\mathcal{E}^{\phi}, \mathcal{F}^{\phi})$, the rooted, oriented, parametrised subordinate Brownian loop measure on $\mathcal{C}^*_X$ is

$$\mu^{*,\phi}_X := \int_0^\infty \frac{dt}{t} \int_X \mathbb{W}^{t,\phi}_{x \to x} \, d\mathrm{vol}_g(x),$$

and its pushforward to $\mathcal{C}_X$ is the unrooted, oriented, unparametrised subordinate Brownian loop measure μ^{ϕ}_X.

2.4.1. *The weighted potential measure.* Since p^{ϕ} is a mixture of $p^{\mathcal{E}}(s, \cdot, \cdot)$ against $\psi^{\phi}_t(ds)$, the loop measure involves a double integral in (t, s). Collapsing the t-integral against the dt/t weight produces a single measure on the subordination variable s. This technique can be found in [SSV12, Chapter 5].

Definition 2.9. The weighted potential measure V_{ϕ} is the σ-finite measure on $(0, \infty)$ defined by

$$\int_{(0,\infty)} h(s) \, V_{\phi}(ds) = \int_0^\infty \frac{dt}{t} \int_{(0,\infty)} h(s) \, \psi^{\phi}_t(ds) \tag{7}$$

for every non-negative measurable h for which the right-hand side is finite. When V_{ϕ} is absolutely continuous we write $V_{\phi}(ds) = V_{\phi}(s) \, ds$; this holds for all our Bernstein functions.

Example 2.10 (Weighted potential measures)**.**

(a) **Brownian motion** ($\phi(\lambda) = \lambda$). Since $\psi^{\phi}_t = \delta_t$, the dt/t integral localises at $s = t$ and $V_{\phi}(ds) = ds/s$.
(b) **Killing** ($\phi(\lambda) = \lambda + \kappa$). Since $\psi^{\phi}_t = e^{-\kappa t}\delta_t$, the same localisation gives $V_{\phi}(ds) = e^{-\kappa s} \, ds/s$.
(c) **α-stable** ($\phi(\lambda) = \lambda^{\alpha/2}$, $\alpha \in (0, 2)$). The subordinator law is $\psi^{\phi}_t(ds) = t^{-2/\alpha} g_{\alpha/2}(s \, t^{-2/\alpha}) \, ds$. Substituting $u = s \, t^{-2/\alpha}$ in (7) gives

$$\int_0^\infty \frac{1}{t} \, t^{-2/\alpha} \, g_{\alpha/2}\big(s \, t^{-2/\alpha}\big) \, dt = \frac{\alpha}{2s} \int_0^\infty g_{\alpha/2}(u) \, du = \frac{\alpha}{2s},$$

since $g_{\alpha/2}$ is a probability density, hence $V_{\phi}(ds) = \frac{\alpha}{2} \frac{ds}{s}$. The multiplicative Haar measure is thus preserved up to the constant $\alpha/2$.

Lemma 2.11. *For all* $x, y \in X$,

$$\int_0^\infty \frac{dt}{t}\, p^\phi(t,x,y) = \int_{(0,\infty)} p^{\mathcal{E}}(s,x,y)\, V_\phi(\,ds). \tag{8}$$

Proof. By (4) and Tonelli,

$$\int_0^\infty \frac{dt}{t}\, p^\phi(t,x,y) = \int_0^\infty \frac{dt}{t} \int_{(0,\infty)} p^{\mathcal{E}}(s,x,y)\, \psi_t^\phi(\,ds),$$

and taking $h(s) = p^{\mathcal{E}}(s,x,y)$ in (7) gives (8). □

3. Decomposition over homotopy classes

Throughout this section, let $\Gamma \subset \mathrm{PSL}(2,\mathbb{R})$ be a torsion-free Fuchsian group, acting freely and properly discontinuously on $\mathbb{H}^2$, and $X = \Gamma\backslash\mathbb{H}^2$ the corresponding geometrically finite hyperbolic surface. The general theory of hyperbolic surfaces can be found in [Bus92, Kat92, Bor07, Rat94]. We write $\rho_{\mathbb{H}^2}$ for the hyperbolic area measure ρ of Section 2.3.3 and ρ_X for the induced area measure on X. Let $(\mathcal{E},\mathcal{F})$ be a Γ-invariant regular symmetric Dirichlet form on $L^2(\mathbb{H}^2,\rho)$ and assume that the semigroup admits a jointly measurable integral kernel $p^{\mathcal{E}}_{\mathbb{H}^2}(t,z,w)$ with respect to ρ, satisfying

$$p^{\mathcal{E}}_{\mathbb{H}^2}(t,z,w) = p^{\mathcal{E}}_{\mathbb{H}^2}(t,hz,hw), \qquad h \in \Gamma,\ t>0,\ z,w \in \mathbb{H}^2.$$

The lifting picture. The quotient map $\pi : \mathbb{H}^2 \to X$ realises $\mathbb{H}^2$ as the universal cover of X, with deck transformation group Γ, the isometries h of $\mathbb{H}^2$ satisfying $\pi \circ h = \pi$. Once a basepoint $x \in X$ and a point $\tilde{x} \in \pi^{-1}(x)$ are fixed, the choice of $\tilde{x}$ determines an isomorphism between the fundamental group $\pi_1(X,x)$ and Γ. Any loop $\omega : [0,t] \to X$ rooted at x admits a unique lift $\tilde{\omega} : [0,t] \to \mathbb{H}^2$ with $\tilde{\omega}(0) = \tilde{x}$, and its endpoint must also lie in $\pi^{-1}(x)$, so $\tilde{\omega}(t) = h_\omega \tilde{x}$ for a unique $h_\omega \in \Gamma$. Thus, the element h_ω records the deck transformation accumulated by ω, and it is the identity if and only if ω is contractible.

Recall that in a free homotopy class the loops are not tied to any basepoint, so different loops start at different points of X and lift to arcs starting at different points of $\mathbb{H}^2$. More specifically, each arc ends at h_ω applied to its own lifted starting point, but the element it records depends on which point of the fibre that arc begins at. E.g., changing the start from $\tilde{x}$ to another point $q\tilde{x}$ carries the whole arc to its q-translate, and the recorded element changes to the conjugate $qh_\omega q^{-1}$, where the displacement length remains the same. As such, free homotopy classes of oriented closed curves on X are in bijection with conjugacy classes in Γ. A free homotopy class is non-trivial if its loops are not null-homotopic and non-peripheral if its loops are neither freely homotopic into a cusp nor freely homotopic to a boundary component. From now on, unless otherwise stated, all free homotopy classes are assumed to be non-trivial and non-peripheral. Each such class singles out a unique closed geodesic representative $m\gamma$ with length $m\ell_\gamma$.

We write $\mathcal{P}_X$ for the set of primitive oriented closed geodesics on X. Each $\gamma \in \mathcal{P}_X$ corresponds to a primitive hyperbolic conjugacy class in Γ with translation length ℓ_γ. Fixing a representative $\tau \in \Gamma$ and conjugating in $\mathrm{PSL}(2,\mathbb{R})$, we may place τ in the standard form

$$\tau : z \longmapsto e^{\ell_\gamma} z, \tag{9}$$

that is, the hyperbolic isometry of $\mathbb{H}^2$ whose axis is the imaginary half-line and whose translation length along that axis is ℓ_γ. For $m \geq 1$, we write $\mathcal{C}_X(\gamma^m)$ for the free homotopy class of oriented closed curves winding m times around γ. This is a non-trivial, non-peripheral free homotopy class corresponding to the conjugacy class of τ^m,

$$[\tau^m]_{\mathrm{conj}} := \{h\tau^m h^{-1} : h \in \Gamma\}.$$

Since Γ is torsion-free and τ is a primitive hyperbolic element, anything commuting with τ^m must preserve the axis of τ, and the elements of Γ preserving that axis form an infinite cyclic subgroup, the centraliser, generated by the primitive translation τ,

$$C_\Gamma(\tau^m) = \langle\tau\rangle = \{\tau^k : k \in \mathbb{Z}\}.$$

We also need to enumerate the elements of $[\tau^m]_{\mathrm{conj}}$ without repetition by identifying when two conjugates coincide. Two elements $h_1, h_2 \in \Gamma$ produce the same conjugate, $h_1\tau^m h_1^{-1} = h_2\tau^m h_2^{-1}$, if and only if $h_1^{-1}h_2 \in C_\Gamma(\tau^m) = \langle\tau\rangle$, that is if and only if h_1 and h_2 lie in the same left coset of $\langle\tau\rangle$ in Γ. Consequently

$$[\tau^m]_{\mathrm{conj}} = \bigsqcup_{r \in \Gamma/\langle\tau\rangle} \{r\tau^m r^{-1}\}, \tag{10}$$

with one distinct conjugate per coset.

Descent to the quotient. Assume $(\mathcal{E}, \mathcal{F})$ is Γ-invariant and its semigroup admits a jointly measurable heat kernel $p^{\mathcal{E}}_{\mathbb{H}^2}$ with $p^{\mathcal{E}}_{\mathbb{H}^2}(t, z, w)$ decaying in $d(z, w)$ fast enough to beat the orbit growth of Γ (this is the case in all our examples). Then for each $t > 0$ the series

$$p^{\mathcal{E}}_X(t, z, w) := \sum_{h \in \Gamma} p^{\mathcal{E}}_{\mathbb{H}^2}(t, \tilde{z}, h\tilde{w}) \tag{11}$$

converges, where $\tilde{z}, \tilde{w} \in \mathbb{H}^2$ are any points with $\pi(\tilde{z}) = z$ and $\pi(\tilde{w}) = w$. We assume throughout that the Γ-invariant form descends to a regular symmetric Dirichlet form on $L^2(X, \rho_X)$ whose transition density is precisely the periodisation (11), and we write $\mu^{\mathcal{E}}_X$ for the associated loop measure. This holds in all subordinate Brownian cases

$$p^{\phi}_X(t, z, w) = \int_{[0,\infty)} p_X(s, z, w)\, \psi^{\phi}_t(\,ds) = \sum_{h \in \Gamma} p^{\phi}_{\mathbb{H}^2}(t, \tilde{z}, h\tilde{w}).$$

The strip domain. In the standard form (9), τ rescales the imaginary part, since $\mathrm{Im}(\tau z) = e^{\ell_\gamma}\,\mathrm{Im}(z)$. Each orbit of $\langle\tau\rangle$ therefore meets the horizontal band $1 \le \mathrm{Im}(z) < e^{\ell_\gamma}$ in exactly one point, and we define the fundamental strip

$$F_\tau := \{z \in \mathbb{H}^2 : 1 \le \mathrm{Im}(z) < e^{\ell_\gamma}\}, \tag{12}$$

which is a fundamental region for $\langle\tau\rangle$ acting on $\mathbb{H}^2$.

Remark 3.1 (Homotopy classes for jump processes). Let $\alpha \in (0, 2)$ and $\phi(\lambda) = \lambda^{\alpha/2}$, so that the subordinate process is the α-stable one. Because it is purely discontinuous, so a sample loop is a càdlàg map into X, it does not have a free homotopy class and does not admit a canonical lift via the path-lifting theorem. For jump processes we therefore define

$$\mu^{\mathcal{E}}_X(\mathcal{C}_X(\gamma^m)) := \int_0^\infty \frac{dt}{t} \int_X \sum_{h \in [\tau^m]_{\mathrm{conj}}} p^{\mathcal{E}}_{\mathbb{H}^2}(t, \tilde{z}, h\tilde{z})\; d\rho_X(z), \tag{13}$$

the part of the loop measure obtained by restricting the periodisation (11) to the conjugacy class $[\tau^m]_{\mathrm{conj}}$.

Despite being discontinuous, the α-stable process can have a path-space interpretation, provided one works on the space carrying the pair (B, S), where the subordinate process is written

$Y_u = B_{S_u}$ with B a Brownian motion and S an independent α-stable subordinator, rather than on the time-changed path Y alone. Conditional on $S_t = s$, the term indexed by h corresponds upstairs to a Brownian bridge from some $\tilde{w} \in \mathbb{H}^2$ to $h\tilde{w}$, and the projection of the full Brownian arc $B|_{[0,s]}$ is a continuous loop with monodromy h, whose conjugacy class $[h]_{\text{conj}}$ is independent of the lift. The subordinator decides only which portions of this arc are observed. At a jump time u, the segment $B|_{[S_{u-},S_u]}$ is deleted, but its endpoints, and hence the accumulated deck transformation, are unchanged. Restricting the periodisation to $[\tau^m]_{\text{conj}}$ therefore selects exactly those marked loops whose underlying Brownian arc lies in $\mathcal{C}_X(\gamma^m)$.

Unfortunately the class is not recoverable from the càdlàg path alone, since the deleted segments may be filled by continuous paths whose projections have different monodromies. It would be interesting to see an intrinsic geometrisation of a càdlàg loop, for example by assigning a canonical continuous interpolation to each jump, but we do not believe this would preserve the closed-form mass formulas below.

Theorem 3.2 (General homotopy class decomposition for hyperbolic surfaces)**.** *Let $\gamma \in \mathcal{P}_X$ be a primitive closed geodesic with hyperbolic representative τ as in (9) and winding number $m \geq 1$. The mass of the Dirichlet form loop measure in a free homotopy class is*

$$\mu_X^{\mathcal{E}}(\mathcal{C}_X(\gamma^m)) = \int_0^\infty \frac{dt}{t} \int_{F_\tau} p_{\mathbb{H}^2}^{\mathcal{E}}(t, z, \tau^m z)\, d\rho_{\mathbb{H}^2}(z). \tag{14}$$

Proof. Step 1: Isolating the conjugacy class. For a continuous process the lifting picture gives the bridge decomposition

$$\mathbb{W}^{t,\mathcal{E}}_{z\to z,\, X} = \sum_{h\in\Gamma} \pi_*\, \mathbb{W}^{t,\mathcal{E}}_{\tilde{z}\to h\tilde{z},\, \mathbb{H}^2}, \tag{15}$$

which decomposes loops rooted at z according to the deck transformation recorded by their lifts, so restricting to loops in $\mathcal{C}_X(\gamma^m)$ corresponds to restricting the sum in (11) (with $\tilde{z} = \tilde{w}$) to $h \in [\tau^m]_{\text{conj}}$; for a jump process this restriction is the definition (13) of Remark 3.1. We have

$$\mu_X^{\mathcal{E}}(\mathcal{C}_X(\gamma^m)) = \int_0^\infty \frac{dt}{t} \int_X \sum_{h\in[\tau^m]_{\text{conj}}} p_{\mathbb{H}^2}^{\mathcal{E}}(t, \tilde{z}, h\tilde{z})\, d\rho_X(z). \tag{16}$$

Step 2: Unfolding to the fundamental strip. Let $F \subset \mathbb{H}^2$ be a fundamental region for Γ. Since $\int_X(\cdot)\, d\rho_X = \int_F(\cdot)\, d\rho_{\mathbb{H}^2}$, the right-hand side of (16) becomes

$$\int_0^\infty \frac{dt}{t} \int_F \sum_{h\in[\tau^m]_{\text{conj}}} p_{\mathbb{H}^2}^{\mathcal{E}}(t, z, hz)\, d\rho_{\mathbb{H}^2}(z). \tag{17}$$

We now unfold the sum over the conjugacy class using (10). For each coset representative $r \in \Gamma/\langle\tau\rangle$, Γ-invariance of the heat kernel gives

$$p_{\mathbb{H}^2}^{\mathcal{E}}(t, z, r\tau^m r^{-1} z) = p_{\mathbb{H}^2}^{\mathcal{E}}(t, r^{-1}z, \tau^m r^{-1} z),$$

and the substitution $w = r^{-1}z$ gives

$$\int_F p_{\mathbb{H}^2}^{\mathcal{E}}(t, z, r\tau^m r^{-1} z)\, d\rho_{\mathbb{H}^2}(z) = \int_{r^{-1}F} p_{\mathbb{H}^2}^{\mathcal{E}}(t, w, \tau^m w)\, d\rho_{\mathbb{H}^2}(w).$$

Summing over the coset representatives,

$$\sum_{r\in\Gamma/\langle\tau\rangle} \int_{r^{-1}F} p_{\mathbb{H}^2}^{\mathcal{E}}(t, w, \tau^m w)\, d\rho_{\mathbb{H}^2}(w) = \int_{\bigcup_r r^{-1}F} p_{\mathbb{H}^2}^{\mathcal{E}}(t, w, \tau^m w)\, d\rho_{\mathbb{H}^2}(w). \tag{18}$$

Since F is a fundamental region for Γ and $\Gamma = \bigsqcup_r \langle\tau\rangle r^{-1}$, the union $\bigsqcup_r r^{-1}F$ is a fundamental region for $\langle\tau\rangle$. Moreover, since τ commutes with τ^m and the kernel is Γ-invariant, $p^{\mathcal{E}}_{\mathbb{H}^2}(t,z,\tau^m z)$ is $\langle\tau\rangle$-invariant, so its integral over any fundamental region of $\langle\tau\rangle$ is the same. We can therefore replace the union by the strip F_τ from (12), giving

$$\mu^{\mathcal{E}}_X(\mathcal{C}_X(\gamma^m)) = \int_0^\infty \frac{dt}{t}\int_{F_\tau} p^{\mathcal{E}}_{\mathbb{H}^2}(t,z,\tau^m z)\, d\rho_{\mathbb{H}^2}(z). \qquad \square$$

Remark 3.3 (Relation to the path integral). The loop measure decomposes over free homotopy classes, and the non-trivial, non-peripheral classes contribute $\sum_{\gamma,m}\mu^{\mathcal{E}}_X(\mathcal{C}_X(\gamma^m))$ where each term is evaluated by Theorem 3.2. Heuristically, this decomposition can be thought of as an interpretation of the homotopy theorem for path integration, which writes a transition amplitude on a topological space X as

$$K = \sum_{[\alpha]\in\pi_1(X,x)} \chi(\alpha)\, K^\alpha,$$

a sum over homotopy classes of a partial amplitude K^α from paths in the class $[\alpha]$, weighted by a unitary character χ of $\pi_1(X,x)$. In our case the masses $\mu^{\mathcal{E}}_X(\mathcal{C}_X(\gamma^m))$ play the role of Wick-rotated partial amplitudes, conjugacy classes are used instead of based classes (since the loops are unrooted), and we have the trivial character $\chi = 1$.

On multiply connected spaces such as hyperbolic surfaces of genus g, different choices of the character χ give physically different theories. The corresponding idea would be to twist the loop measure by a flat unitary bundle on X (equivalently a unitary representation of $\pi_1(X)$), which replaces (11) by its twisted form; this is done in later sections. In the case of graphs and compact surfaces, see [LJ24] for various physical interpretations of twisted loop measures and [PS25] for a detailed mathematical construction.

3.1. **Subordinate Brownian cases.** We now specialise to subordinate Brownian loop measure. Applying Theorem 3.2 with $p^{\mathcal{E}}_{\mathbb{H}^2} = p^{\phi}_{\mathbb{H}^2}$ and expanding via (6) gives

$$\mu^{\phi}_X(\mathcal{C}_X(\gamma^m)) = \int_0^\infty \frac{dt}{t}\int_{F_\tau}\int_{[0,\infty)} p_{\mathbb{H}^2}(s,z,\tau^m z)\,\psi^{\phi}_t(\,ds)\, d\rho_{\mathbb{H}^2}(z). \tag{19}$$

The strategy is to evaluate the spatial integral using an identity for the Brownian heat kernel on $\mathbb{H}^2$, then collapse the dt/t-integral into the weighted potential measure V_ϕ. We also write $L := m\ell_\gamma$ throughout.

Lemma 3.4 (Wang–Xue [WX25])**.** *For any* $s > 0$ *and* $m \geq 1$,

$$\int_{F_\tau} p_{\mathbb{H}^2}(s,z,e^L z)\, d\rho_{\mathbb{H}^2}(z) = \frac{\ell_\gamma}{2\sinh(L/2)}\cdot\frac{e^{-s/4}\,e^{-L^2/(4s)}}{2\sqrt{\pi s}}. \tag{20}$$

Theorem 3.5 (Mass of subordinate Brownian loop measure on hyperbolic surfaces)**.** *Let* ϕ *be a Bernstein function satisfying Assumption 2.3. Then for any* $\gamma \in \mathcal{P}_X$ *and* $m \geq 1$,

$$\mu^{\phi}_X(\mathcal{C}_X(\gamma^m)) = \frac{\ell_\gamma}{2\sinh(L/2)}\int_0^\infty \frac{e^{-s/4}\,e^{-L^2/(4s)}}{2\sqrt{\pi s}}\, V_\phi(\,ds), \tag{21}$$

where $L = m\ell_\gamma$ *and* V_ϕ *is the weighted potential measure of Definition 2.9.*

Proof. Starting from (19), the inner spatial integral is evaluated by Lemma 3.4

$$\int_{F_\tau} p_{\mathbb{H}^2}(s,z,\tau^m z)\, d\rho_{\mathbb{H}^2}(z) = \frac{\ell_\gamma}{2\sinh(L/2)}\cdot\frac{e^{-s/4}\,e^{-L^2/(4s)}}{2\sqrt{\pi s}}.$$

Substituting into (19) gives

$$\mu_X^\phi(\mathcal{C}_X(\gamma^m)) = \frac{\ell_\gamma}{2\sinh(L/2)} \int_0^\infty \frac{dt}{t} \int_{[0,\infty)} \frac{e^{-s/4}\, e^{-L^2/(4s)}}{2\sqrt{\pi s}}\, \psi_t^\phi(\,ds). \tag{22}$$

Applying Lemma 2.11 with $h(s) = \frac{e^{-s/4}\, e^{-L^2/(4s)}}{2\sqrt{\pi s}}$ collapses the double integral into a single integral against V_ϕ, giving (21). □

Definition 3.6. For the following we isolate the integral against the weighted potential measure

$$\mathcal{I}_\phi(L) := \int_0^\infty \frac{e^{-s/4}\, e^{-L^2/(4s)}}{2\sqrt{\pi s}}\, V_\phi(\,ds), \qquad L > 0, \tag{23}$$

so that Theorem 3.5 reads

$$\mu_X^\phi(\mathcal{C}_X(\gamma^m)) = \frac{\ell_\gamma}{2\sinh(L/2)}\, \mathcal{I}_\phi(L), \qquad L = m\ell_\gamma. \tag{24}$$

3.1.1. *Recovery of the Brownian case.* For $\phi(\lambda) = \lambda$ we have $V_\phi(\,ds) = ds/s$, and (23) becomes

$$\mathcal{I}_{BM}(L) = \int_0^\infty \frac{e^{-s/4}\, e^{-L^2/(4s)}}{2\sqrt{\pi}\, s^{3/2}}\, ds.$$

Using the integral identity $\int_0^\infty s^{-3/2}\, e^{-as-b/s}\, ds = \sqrt{\frac{\pi}{b}}\, e^{-2\sqrt{ab}}$, we take $a = 1/4$ and $b = L^2/4$. The right-hand side is $\frac{2\sqrt{\pi}}{L}\, e^{-L/2}$ so

$$\mu_X(\mathcal{C}_X(\gamma^m)) = \frac{\ell_\gamma}{2\sinh(L/2)} \cdot \frac{1}{2\sqrt{\pi}} \cdot \frac{2\sqrt{\pi}}{L}\, e^{-L/2} = \frac{\ell_\gamma}{L} \cdot \frac{e^{-L/2}}{2\sinh(L/2)} = \frac{1}{m} \cdot \frac{1}{e^L - 1},$$

since $\ell_\gamma/L = 1/m$ and $e^{-L/2}/(2\sinh(L/2)) = 1/(e^L - 1)$. This recovers the formula of Wang–Xue [WX25, Lemma 3.2].

3.1.2. *Brownian with killing.* For $\phi(\lambda) = \lambda + \kappa$ with $\kappa \geq 0$, the weighted potential measure is $V_\phi(\,ds) = e^{-\kappa s}\, ds/s$, and hence

$$\mathcal{I}_\kappa(L) = \int_0^\infty \frac{e^{-(1/4+\kappa)s}\, e^{-L^2/(4s)}}{2\sqrt{\pi}\, s^{3/2}}\, ds.$$

Applying the integral identity with $a = \frac{1}{4} + \kappa$ and $b = \frac{L^2}{4}$ gives

$$\mathcal{I}_\kappa(L) = \frac{e^{-L\sqrt{1/4+\kappa}}}{L}, \tag{25}$$

and therefore, writing $\mu_X^\kappa = \mu_X^\phi$ for $\phi(\lambda) = \lambda + \kappa$

$$\mu_X^\kappa(\mathcal{C}_X(\gamma^m)) = \frac{\ell_\gamma}{2\sinh(L/2)} \cdot \frac{e^{-L\sqrt{1/4+\kappa}}}{L} = \frac{1}{m} \cdot \frac{e^{(\frac{1}{2} - \sqrt{\frac{1}{4}+\kappa})\,L}}{e^L - 1}. \tag{26}$$

This is exactly Lemonde–Wang [LW26, Lemma 3.1]. Setting $\kappa = 0$ recovers the Brownian formula.

Remark 3.7 (The range $\kappa \geq -\frac{1}{4}$). For $\kappa > 0$, μ_X^κ is the mass of the Brownian loop measure with a constant killing rate where the operator of the process is no longer the Laplacian but Schrödinger with a constant potential $+\kappa$. For $\kappa \in [-\frac{1}{4}, 0)$ the function $\phi(\lambda) = \lambda + \kappa$ is no longer Bernstein, since its killing rate is negative, but the formula (26) continues to make sense analytically. Writing $s = \frac{1}{2} + \sqrt{\frac{1}{4} + \kappa}$, where s here denotes the spectral parameter, the condition

$\kappa \geq -\frac{1}{4}$ is what keeps s real, with $\kappa = -\frac{1}{4}$ giving $s = \frac{1}{2}$, the bottom of the L^2-spectrum of $\Delta_{\mathbb{H}^2}$, and the integral (25) converges throughout. For more general killing rates see Section 3.2.

3.1.3. *α-stable case.* For $\phi(\lambda) = \lambda^{\alpha/2}$ with $\alpha \in (0,2)$, the weighted potential measure is $V_\phi(\,ds) = \frac{\alpha}{2}\,ds/s$, so the integral in (23) is

$$\mathcal{I}_\alpha(L) = \frac{\alpha}{2}\,\mathcal{I}_{BM}(L) = \frac{\alpha}{2}\cdot\frac{e^{-L/2}}{L}, \qquad L > 0. \tag{27}$$

Hence the mass of the α-stable loop measure μ_X^α (interpreted as in Remark 3.1) is

$$\mu_X^\alpha(\mathcal{C}_X(\gamma^m)) = \frac{\alpha}{2}\,\mu_X(\mathcal{C}_X(\gamma^m)).$$

This identity is unsurprising given scale-invariance, namely that the self-similar subordinators are precisely the stable ones and the weight dt/t is itself scale-invariant. When the two are combined in (7), the only measure on $(0,\infty)$ compatible with both scalings is a constant multiple of ds/s. Unfortunately this collapse gives us no extra information about the geometry of X.

Hence to get a decomposition that differs from the Brownian one in form one must break scale-invariance; this could be done, for example, by subordinating the shifted Laplacian $\Delta_{\mathbb{H}^2} + \kappa$.

3.1.4. *Shifted α-stable case.* For $\phi(\lambda) = (\lambda+\kappa)^{\alpha/2}$ with $\alpha \in (0,2)$ and $\kappa \geq 0$, the operator is $(\Delta_{\mathbb{H}^2} + \kappa)^{\alpha/2}$; this ϕ is a complete Bernstein function, being the composition of $\lambda \mapsto \lambda + \kappa$ with $u \mapsto u^{\alpha/2}$, so the framework of Section 2 applies.

Its subordinator law is an exponential tilt of the α-stable law

$$e^{-t(\lambda+\kappa)^{\alpha/2}} = \int_0^\infty e^{-u(\lambda+\kappa)}\,\eta_t^\alpha(u)\,du = \int_0^\infty e^{-\lambda s}\,e^{-\kappa s}\,\eta_t^\alpha(s)\,ds,$$

we have $\psi_t^\phi(\,ds) = e^{-\kappa s}\,\eta_t^\alpha(s)\,ds$. Substituting into (7) and applying the collapse to $h(s)\,e^{-\kappa s}$ gives

$$V_\phi(\,ds) = \frac{\alpha}{2}\,e^{-\kappa s}\,\frac{ds}{s}. \tag{28}$$

Consequently

$$\mathcal{I}_\phi(L) = \frac{\alpha}{2}\int_0^\infty \frac{e^{-(1/4+\kappa)s}\,e^{-L^2/(4s)}}{2\sqrt{\pi}\,s^{3/2}}\,ds = \frac{\alpha}{2}\cdot\frac{e^{-L\sqrt{1/4+\kappa}}}{L},$$

by the integral identity with $a = \frac{1}{4}+\kappa$ and $b = \frac{L^2}{4}$, and (24) gives

$$\mu_X^\phi(\mathcal{C}_X(\gamma^m)) = \frac{\ell_\gamma}{2\sinh(L/2)}\cdot\frac{\alpha}{2}\cdot\frac{e^{-L\sqrt{1/4+\kappa}}}{L} = \frac{\alpha}{2}\cdot\frac{1}{m}\cdot\frac{e^{(\frac{1}{2}-\sqrt{\frac{1}{4}+\kappa})\,L}}{e^L-1}. \tag{29}$$

The left-hand side is interpreted as in Remark 3.1.

3.2. **A quantum mechanical digression.** It is well known from diffusion theory, or more specifically Euclidean quantum mechanics, that ordinary Brownian motion corresponds, after Wick rotation, to a free non-relativistic quantum particle. But the more interesting connections to quantum theory arise when a potential is added to the free Hamiltonian, and in our case the potential is the constant killing.

Recall that the wave function ψ of an isolated system satisfies, with our sign convention, the Schrödinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi, \qquad \hat{H} = \frac{\hbar^2}{2m}\Delta_X + V(x),$$

where $\hat{H}$ is the Hamiltonian, $V(x) \geq 0$ is the potential energy at $x \in X$. The solution is the unitary group $\psi(x,t) = e^{-it\hat{H}/\hbar}\psi(x,0)$, unitary since $\hat{H}$ is self-adjoint. Notice that the propogator (i.e., transition amplitude) is then $K(t,x,y) = \langle y|e^{-it\hat{H}/\hbar}|x\rangle$, which we had for homotopy classes in Remark 3.3. Under the Wick rotation $t = -i\tau$ the unitary group becomes the contraction semigroup $e^{-\tau\hat{H}/\hbar}$, which is well-defined for $\tau \geq 0$ since $\hat{H} \geq 0$; the Euclidean time τ is precisely the diffusion time of Section 2, so we write t for it from now on. In units where $\hbar^2/2m = 1$, and absorbing $\hbar$ into the potential, the semigroup has integral kernel $p^V(t,x,y)$, which is the transition density of Brownian motion killed at rate V. The Feynman–Kac formula gives

$$p^V(t,x,y) = \int_{C([0,t];X)} e^{-\int_0^t V(\omega(r))\,dr}\,\mathbb{W}^t_{x\to y}(\,d\omega),$$

with each Brownian bridge weighted by its probability of surviving the killing along its trajectory. For $V = \kappa$ the weight is $e^{-\kappa t}$, which gives $p^\kappa(t,x,y) = e^{-\kappa t}p(t,x,y)$ and thus recovers Example 2.6. So μ^κ_X is the intensity of the loop ensemble of a Euclidean quantum particle in a constant potential, namely the intensity of the loop soup in Section 3.3. Also note that our Feynman-Kac formula is the rigorous form of the Euclidean path integral $\int \mathcal{D}\omega\, e^{-S[\omega]}$ with action $S[\omega] = \int_0^t \big(\frac{1}{4}|\dot{\omega}(r)|^2 + V(\omega(r))\big)\,dr$, where the kinetic part of the action is included into the bridge measure $\mathbb{W}^t_{x\to y}$ and the potential part is the weight; $\mathcal{D}\omega$ is the formal Lebesgue measure on continuous paths, which does not exist as a measure.

Looking ahead, this link with quantum particles also gives a physical motivation for the appearance of Brownian loops with killing in zeta-regularised determinants of the Laplacian in Section 5. In Euclidean quantum field theory, consider a free real scalar field φ of mass m on the hyperbolic surface X, with $\kappa = m^2 > 0$ and effective action

$$S_E[\varphi] = \frac{1}{2}\int_X \big(|\nabla\varphi|^2 + \kappa\,\varphi^2\big)\,d\mathrm{vol}_g = \frac{1}{2}\,\langle \varphi, (\Delta_X + \kappa)\varphi\rangle,$$

where the second equality uses that Δ_X is the positive Laplacian. Since the action is quadratic, the Euclidean path integral is Gaussian and gives

$$\mathcal{Z}^\kappa_X = \int \mathcal{D}\varphi\; e^{-S_E[\varphi]} \;\propto\; \det(\Delta_X + \kappa)^{-1/2}.$$

Consequently the one-loop effective action of the real scalar field is

$$\Gamma^{(1)}_X(\kappa) = -\log \mathcal{Z}^\kappa_X = \frac{1}{2}\,\log\det(\Delta_X + \kappa).$$

Thus the determinant may be expressed through the heat semigroup by the Schwinger propertime representation

$$-\log\det(\Delta_X + \kappa) = \int_0^\infty \frac{dt}{t}\,e^{-\kappa t}\;\mathrm{Tr}\big(e^{-t\Delta_X}\big),$$

where in Section 5 we explain the details. When $e^{-t\Delta_X}$ is trace class,

$$\mathrm{Tr}\big(e^{-t\Delta_X}\big) = \int_X p(t,x,x)\;d\mathrm{vol}_g(x),$$

the heat trace is built from Brownian paths which start and return to the same point. Integration over t with the Haar weighting dt/t, together with the killing weight $e^{-\kappa t}$, is precisely the structure of the Brownian loop measure with killing, and the Schwinger proper-time reads as

$$-\log\det(\Delta_X+\kappa)=|\mu_X^\kappa|_{\mathrm{reg}},$$

the total mass of the loop measure with killing over all loops, including the regularised divergent contribution of the contractible class (and of the peripheral classes, when present). Hence

$$\mathcal{Z}_X^\kappa \;\propto\; \exp\Big(\tfrac{1}{2}\,|\mu_X^\kappa|_{\mathrm{reg}}\Big),$$

so the partition function of a free real scalar field of mass $\sqrt{\kappa}$ is, up to normalisation, the exponential of half the regularised total mass of Brownian loops on X with killing rate κ; the factor $\frac{1}{2}$ is the power $\det^{-1/2}$ of a single real field.

3.3. **The loop soup and its Poissonian structure.** Any σ-finite measure can serve as the intensity of a Poisson point process, and thus the mass of subordinate Brownian loop measure in a free homotopy class can be interpreted as the mean of a Poisson random variable, which lets us speak of the distribution of topological quantities rather than only their expectations.

Following Lawler and Werner [LW04] (see also [LJ11]), let ϕ be a Bernstein function satisfying Assumption 2.3, and for $c>0$ let $\mathcal{L}_c$ be the Poisson point process of loops on X with intensity measure $c\,\mu_X^\phi$. Thus $\mathcal{L}_c$ is a random countable collection of loops, the subordinate Brownian loop soup, and for any measurable set A of loops with $\mu_X^\phi(A)<\infty$ the number

$$N_A := \#\{\eta\in\mathcal{L}_c : \eta\in A\}$$

is a Poisson random variable with mean $c\,\mu_X^\phi(A)$. The parameter c is the intensity of the soup.

When the process has jumps, the class $\mathcal{C}_X(\gamma^m)$ is not a measurable set of càdlàg loops by Remark 3.1. In that case we take $\mathcal{L}_c$ to be the Poisson point process of marked loops carrying the pair (B,S), on which the monodromy class is measurable and has the same intensity $c\,\mu_X^\phi(\mathcal{C}_X(\gamma^m))$. Applying this to the homotopy classes gives:

Proposition 3.8 (Poissonian structure of homotopy classes)**.** *For $\gamma\in\mathcal{P}_X$ and $m\geq 1$, the number of loops of $\mathcal{L}_c$ in the free homotopy class $\mathcal{C}_X(\gamma^m)$ (for jump processes, of marked loops, as in Remark 3.1) is a Poisson random variable of mean $c\,\mu_X^\phi(\mathcal{C}_X(\gamma^m))$, and for any finite collection of pairwise distinct, hence disjoint, classes these variables are jointly independent.*

Proof. Distinct free homotopy classes are disjoint measurable sets, so both statements are immediate from the properties of a Poisson point process. □

3.4. **Length spectra identities.** The two fundamental properties of Brownian loop measure, restriction and conformal invariance, give an identity between geodesic length spectra of different surfaces. Removing a polar set P from X and equipping the surface $X\setminus P$ with its own complete hyperbolic metric leaves the mass of Brownian loop measure unchanged, and comparing the two sides relates the length spectra of the two surfaces. For subordinate Brownian loop measures, in the diffusion cases (where homotopy classes make sense), only a trivial form of the identity survives.

From the potential theory of Markov processes (see, for example, [BG68]), a Borel set $P\subset X$ is polar for a given process if, from every starting point, the process almost surely never hits P at a positive time. For Brownian motion on a Riemann surface this holds exactly when P has zero logarithmic capacity in every local chart, and in particular every singleton is polar. Polar sets form a σ-ideal, being closed under subsets and countable unions, so every countable set is polar. A killing rate does not change the paths of the process, so for $\phi(\lambda) = \lambda + \kappa$ the polar sets are again those of Brownian motion. We take P to be a closed discrete set, hence countable and polar.

Consequently μ_X^κ is supported on loops avoiding P, and the restriction property gives

$$\mu^\kappa_{X,\,g}(\mathcal{C}_X(\gamma^m)) = \mu^\kappa_{X\setminus P,\,g}(\mathcal{C}_X(\gamma^m)),$$

where we make the dependence on the metric explicit. Here g on the right means the ambient metric restricted to $X \setminus P$, not the complete hyperbolic metric of the punctured surface, and the class $\mathcal{C}_X(\gamma^m)$ is measured in X on both. The mass of the Brownian with killing loop measure is thus unchanged by puncturing along P, so long as the metric g itself is unchanged.

This is significantly weaker than the Brownian case. There conformal invariance allows one to replace g on $X \setminus P$ by the unique complete hyperbolic metric g' of $X \setminus P$, a different metric with a cusp at each point of P, and the identity becomes a comparison of geodesic length spectra between the two surfaces. For a subordinate process the identity no longer holds; recall that in 2D a conformal change $g' = e^{2\sigma} g$ rescales the Laplacian by $\Delta_{X,g'} = e^{-2\sigma}\Delta_{X,g}$, and this conformal covariance rescaling is not preserved by ϕ, since $\phi(e^{-2\sigma}\Delta_{X,g}) \neq e^{-2\sigma}\phi(\Delta_{X,g})$ unless $\phi(\lambda) = c\lambda$ for some constant $c > 0$. We recall the Brownian result below.

Theorem 3.9 (Wang–Xue [WX25, Theorem 4.2])**.** *Let X be a complete hyperbolic surface without boundary, possibly of infinite type and possibly with cusps or funnels, and let $P \subset X$ be a non-empty, closed, polar set. Let $X' = X \setminus P$ equipped with its unique complete hyperbolic metric. Then for any $\gamma \in \mathcal{P}_X$ and $m \geq 1$,*

$$\frac{1}{m}\cdot\frac{1}{e^{m\ell_\gamma}-1} = \sum_{\substack{\gamma'\in\mathcal{P}_{X'},\, m'\geq 1\\ \gamma'^{m'}\simeq_X \gamma^m}} \frac{1}{m'}\cdot\frac{1}{e^{m'\ell_{\gamma'}}-1},$$

where $\simeq_X$ denotes free homotopy as curves in X, and ℓ_γ and $\ell_{\gamma'}$ are the primitive geodesic lengths measured on X and X' respectively.

3.4.1. *Marked length spectrum from Brownian loop measure.* Another identity shows that the mass of subordinate Brownian loop measure recovers the marked length spectrum of X in the case of closed surfaces.

Recall that every free homotopy class of closed curves on X is $\mathcal{C}_X(\gamma^m)$ for a unique $\gamma \in \mathcal{P}_X$ and $m \geq 1$, and write $\ell_g(\eta)$ for the length of a loop η measured in the metric g.

Definition 3.10 (Marked length spectrum)**.** The marked length spectrum of (X, g) is the function defined on the set of non-trivial free homotopy classes of closed curves on X,

$$\mathrm{MLS}\colon \mathcal{C}_X(\gamma^m) \longmapsto \inf_{\eta\in\mathcal{C}_X(\gamma^m)} \ell_g(\eta),$$

which assigns to each free homotopy class the infimum of the lengths in the class. On a hyperbolic surface the infimum is attained by the unique closed geodesic in the class, so $\mathrm{MLS}(\mathcal{C}_X(\gamma^m)) = m\ell_\gamma$.

The marking, by which we mean the record of which class realises which length, is important as there exist non-isometric hyperbolic surfaces whose geodesic lengths agree as a set (see Vignéras [**?**]), so the values of MLS alone do not determine X. Burns–Katok [**?**] conjectured that a negatively curved metric on a closed manifold is determined up to isometry by its marked length spectrum, and in 2D this was proved by Otal [?] and Croke [?].

Proposition 3.11. *For every $\gamma \in \mathcal{P}_X$,*

$$\ell_\gamma = \log\left(1 + \frac{1}{\mu_X(\mathcal{C}_X(\gamma))}\right). \tag{30}$$

For $\phi(\lambda) = \lambda + \kappa$ with $\kappa \geq -\frac{1}{4}$, the mass $\mu_X^\kappa(\mathcal{C}_X(\gamma))$ is a strictly decreasing function of ℓ_γ and hence again determines it. Both statements hold for every $m \geq 1$, so in either case the loop masses determine MLS*.*

Proof. By Section 3.1.1, $\mu_X(\mathcal{C}_X(\gamma)) = 1/(e^{\ell_\gamma} - 1)$, and solving for ℓ_γ gives (30). By (26), $\mu_X^\kappa(\mathcal{C}_X(\gamma)) = e^{(\frac{1}{2}-\sqrt{\frac{1}{4}+\kappa})\ell_\gamma}/(e^{\ell_\gamma} - 1)$, whose logarithmic derivative in ℓ_γ is

$$\left(\tfrac{1}{2} - \sqrt{\tfrac{1}{4} + \kappa}\right) - \frac{e^{\ell_\gamma}}{e^{\ell_\gamma} - 1} < \tfrac{1}{2} - 1 < 0.$$

For general m the same computations apply. $\square$

Corollary 3.12 (Loop masses determine the hyperbolic surface)**.** *Let X be a closed hyperbolic surface, let g_1, g_2 be hyperbolic metrics on X, and fix $\kappa \geq -\frac{1}{4}$. If*

$$\mu^\kappa_{X,\,g_1}(\mathcal{C}_X(\gamma^m)) = \mu^\kappa_{X,\,g_2}(\mathcal{C}_X(\gamma^m))$$

for every free homotopy class $\mathcal{C}_X(\gamma^m)$, then (X, g_1) and (X, g_2) are isometric by an isometry isotopic to the identity, and hence define the same point in the Teichmüller space.

Proof. By Proposition 3.11 the mass in a free homotopy class determines the length of its geodesic representative, so the hypothesis gives that g_1 and g_2 have the same marked length spectrum, with the identity marking. Hyperbolic metrics are negatively curved, so by Otal [?] and Croke [?] there is an isometry between g_1 and g_2 homotopic, and hence isotopic, to the identity, so the two metrics define the same point of the Teichmüller space. $\square$

4. Zeta Functions and the Total Mass of Brownian Loops

4.1. **Selberg zeta identities.** In this subsection we show that the mass of Brownian loop measure, summed over all free homotopy classes, gives a family of zeta function identities. Throughout, total mass refers to this sum over non-trivial, non-peripheral classes; the full loop measure always has infinite mass owing to the trivial class.

Definition 4.1 (Selberg zeta function)**.** The Selberg zeta function of a geometrically finite hyperbolic surface $X = \Gamma\backslash\mathbb{H}^2$ is defined as the double Euler product

$$Z_X(s) := \prod_{\gamma\in\mathcal{P}_X} \prod_{k=0}^{\infty} \left(1 - e^{-(s+k)\ell_\gamma}\right), \qquad \mathrm{Re}(s) > \delta, \tag{31}$$

where δ is the critical exponent of Γ. It converges absolutely for $\mathrm{Re}(s) > \delta$ and admits a meromorphic continuation to $\mathbb{C}$.

The critical exponent δ is the exponent of convergence of the Poincaré series $\sum_{h\in\Gamma} e^{-s\, d(z,h\cdot z)}$, measuring the rate at which the orbit Γz accumulates in $\mathbb{H}^2$. The same rate governs the proliferation of closed geodesics through the prime geodesic theorem (see Subsection 4.2). Over the past decades several more interpretations of the exponent have been discovered. By Patterson–Sullivan theory it equals the Hausdorff dimension of the limit set $\Lambda(\Gamma) \subset \partial\mathbb{H}^2$, and it equals the topological entropy of the geodesic flow on the non-wandering set $\Omega \subset T^1X$ in the unit tangent bundle of X. When $\delta > \frac{1}{2}$ it also determines the smallest L^2-eigenvalue $\lambda_0 = \delta(1-\delta)$ of Δ_X, lying below $\frac{1}{4}$, the bottom of the continuous spectrum when X is non-compact. In the finite-area case $\delta = 1$, giving $\Omega = T^1X$ and $\lambda_0 = 0$.

Taking the logarithm of (31) and expanding $-\log(1-x) = \sum_{m=1}^{\infty} x^m/m$ gives, for $\mathrm{Re}(s) > \delta$,

$$-\log Z_X(s) = \sum_{\gamma\in\mathcal{P}_X}\sum_{m=1}^{\infty}\frac{1}{m}\cdot\frac{e^{(1-s)m\ell_\gamma}}{e^{m\ell_\gamma}-1}, \tag{32}$$

where the geometric series gives $\sum_{k=0}^{\infty} e^{-(s+k)m\ell_\gamma} = e^{(1-s)m\ell_\gamma}/(e^{m\ell_\gamma}-1)$.

Lemma 4.2 (Selberg zeta criterion). *Let $\mathcal{I}_\phi$ be as in Definition 3.6 and suppose there exist a constant $C > 0$ and a real number $s > \delta$, both independent of L, such that*

$$\frac{L}{2\sinh(L/2)}\mathcal{I}_\phi(L) = C\cdot\frac{e^{(1-s)L}}{e^L-1}, \qquad L > 0. \tag{33}$$

Then

$$\sum_{\gamma\in\mathcal{P}_X}\sum_{m=1}^{\infty}\mu_X^\phi(\mathcal{C}_X(\gamma^m)) = -C\,\log Z_X(s). \tag{34}$$

Proof. By (24) and $L = m\ell_\gamma$,

$$\mu_X^\phi(\mathcal{C}_X(\gamma^m)) = \frac{\ell_\gamma}{2\sinh(L/2)}\mathcal{I}_\phi(L) = \frac{1}{m}\cdot\frac{L}{2\sinh(L/2)}\mathcal{I}_\phi(L).$$

Substituting (33) gives $\mu_X^\phi(\mathcal{C}_X(\gamma^m)) = C\cdot\frac{1}{m}\cdot e^{(1-s)L}/(e^L-1)$, and summing over $\gamma\in\mathcal{P}_X$ and $m \geq 1$ matches the right-hand side of (32). Absolute convergence is ensured by $s > \delta$. ☐

4.1.1. *The killing case.* For $\phi(\lambda) = \lambda + \kappa$, formula (25) gives $\mathcal{I}_\kappa(L) = e^{-L\sqrt{1/4+\kappa}}/L$, so

$$\frac{L}{2\sinh(L/2)}\mathcal{I}_\kappa(L) = \frac{e^{-L\sqrt{1/4+\kappa}}}{2\sinh(L/2)} = \frac{e^{(1-s)L}}{e^L-1}, \qquad s = \tfrac{1}{2}+\sqrt{\tfrac{1}{4}+\kappa},$$

giving (33) with $C = 1$.

Corollary 4.3 (Selberg zeta identity, killing case). *For any $\kappa \geq -\frac{1}{4}$ with $s = \frac{1}{2}+\sqrt{\frac{1}{4}+\kappa} > \delta$,*

$$\sum_{\gamma\in\mathcal{P}_X}\sum_{m=1}^{\infty}\mu_X^\kappa(\mathcal{C}_X(\gamma^m)) = -\log Z_X\big(\tfrac{1}{2}+\sqrt{\tfrac{1}{4}+\kappa}\big). \tag{35}$$

Setting $\kappa = 0$ gives $s = 1$, hence the Brownian total mass is $\sum_{\gamma,m}\mu_X(\mathcal{C}_X(\gamma^m)) = -\log Z_X(1)$. When X has infinite area, $\delta < 1$ and this quantity is finite; in the finite-area case $\delta = 1$ and the sum diverges (see Subsection 4.2). The killing identity was originally shown in [LW26].

Remark 4.4 (Bosonic partition function interpretation). The total mass can also be interpreted as the logarithm of a bosonic partition function at zero chemical potential. For $\mathrm{Re}(s) > \delta$ and $\gamma\in\mathcal{P}_X$, set

$$\mathcal{Z}_\gamma(s) := \prod_{k=0}^{\infty}\frac{1}{1-e^{-(s+k)\ell_\gamma}}, \qquad \mathcal{Z}(s) := \prod_{\gamma\in\mathcal{P}_X}\mathcal{Z}_\gamma(s) = Z_X(s)^{-1},$$

where Z_X is the Selberg zeta function. Then Corollary 4.3 reads

$$\sum_{\gamma\in\mathcal{P}_X}\sum_{m=1}^{\infty}\mu_X^\kappa(\mathcal{C}_X(\gamma^m)) = \log\mathcal{Z}(s) = -\log Z_X(s), \qquad s = \tfrac{1}{2}+\sqrt{\tfrac{1}{4}+\kappa}.$$

Each $\mathcal{Z}_\gamma(s)$ is the partition function of a family of bosonic modes indexed by $k \geq 0$ with weights $(s+k)\ell_\gamma$. The product over $\gamma \in \mathcal{P}_X$ converges for $\mathrm{Re}(s) > \delta$, and $\mathcal{Z}(s)$ can be thought of as the grand canonical partition function of a free, non-interacting Bose gas at zero chemical potential.

4.1.2. *A family of zeta function identities.* The link between the total mass of the Brownian loop measure and the Selberg zeta function yields, in principle, an identity for any zeta function built from the length spectrum. The Selberg zeta identity is the most natural of these. E.g., for dynamical zeta functions, such as the Ruelle zeta function or its twisted versions, the corresponding identities are more difficult to use in a meaningful way.

Definition 4.5 (Ruelle zeta function)**.** The Ruelle zeta function of X is

$$R_X(s) := \prod_{\gamma \in \mathcal{P}_X} \left(1 - e^{-s\ell_\gamma}\right), \qquad \mathrm{Re}(s) > \delta. \tag{36}$$

It is related to the Selberg zeta function by

$$R_X(s) = \frac{Z_X(s)}{Z_X(s+1)}, \qquad \text{equivalently} \quad Z_X(s) = \prod_{k=0}^{\infty} R_X(s+k), \tag{37}$$

so the meromorphic continuation of R_X follows from that of Z_X.

The twisted version weights each geodesic by a representation ρ of Γ. Let $\rho\colon \Gamma \to GL(V_\rho)$ be a finite-dimensional complex representation, not necessarily unitary, and for $\gamma \in \mathcal{P}_X$ let τ represent the corresponding primitive hyperbolic conjugacy class. The twisted Ruelle zeta function is

$$R_X(s,\rho) := \prod_{\gamma \in \mathcal{P}_X} \det\left(I - \rho(\tau)\, e^{-s\ell_\gamma}\right), \tag{38}$$

which reduces to $R_X(s)$ when ρ is trivial. The determinant depends only on the conjugacy class of τ so the product is well-defined. It converges absolutely for $\mathrm{Re}(s) > c_\rho$, where c_ρ is governed by the growth rate $\|\rho(\tau)\| \leq C_\rho\, e^{c\,\ell_\gamma}$ of the representation along geodesics [?, Theorem 3.1]; for unitary ρ one may take $c_\rho = \delta$.

Corollary 4.6 (Twisted Ruelle zeta identity)**.** *Let* $\kappa_-(s) := s(s-1)$ *and* $\kappa_+(s) := s(s+1)$*. For* $\mathrm{Re}(s) > \frac{1}{2}$ *the principal square root satisfies*

$$\tfrac{1}{2} + \sqrt{\tfrac{1}{4} + \kappa_-(s)} = s, \qquad \tfrac{1}{2} + \sqrt{\tfrac{1}{4} + \kappa_+(s)} = s+1.$$

Then, for $\mathrm{Re}(s) > \max(c_\rho, \frac{1}{2})$,

$$-\log R_X(s,\rho) = \sum_{\gamma \in \mathcal{P}_X} \sum_{m=1}^{\infty} \mathrm{tr}\ \rho(\tau^m) \left[\mu_X^{\kappa_-(s)}(\mathcal{C}_X(\gamma^m)) - \mu_X^{\kappa_+(s)}(\mathcal{C}_X(\gamma^m))\right] = \sum_{\gamma \in \mathcal{P}_X} \sum_{m=1}^{\infty} \frac{\mathrm{tr}\ \rho(\tau^m)\, e^{-sm\ell_\gamma}}{m}. \tag{39}$$

Proof. Expanding the logarithm of (38) via $-\log\det(I - M) = \sum_{m \geq 1} \mathrm{tr}(M^m)/m$ gives

$$-\log R_X(s,\rho) = \sum_{\gamma \in \mathcal{P}_X} \sum_{m=1}^{\infty} \frac{\mathrm{tr}\ \rho(\tau^m)\, e^{-sm\ell_\gamma}}{m}.$$

By (26),

$$\mu_X^{\kappa_-(s)}\left(\mathcal{C}_X(\gamma^m)\right) - \mu_X^{\kappa_+(s)}\left(\mathcal{C}_X(\gamma^m)\right) = \frac{1}{m} \cdot \frac{e^{(1-s)L} - e^{-sL}}{e^L - 1} = \frac{e^{-sL}}{m},$$

since $e^{(1-s)L} - e^{-sL} = e^{-sL}(e^L - 1)$. Matching term by term gives (39), and absolute convergence for $\mathrm{Re}(s) > \max(c_\rho, \frac{1}{2})$ follows from that of the product. $\square$

Passing from $\kappa_-(s)$ to $\kappa_+(s)$ suppresses longer loops more strongly, and the difference $\mu_X^{\kappa_-} - \mu_X^{\kappa_+}$ isolates the net contribution of each homotopy class between the two rates.

4.2. **Finiteness of total mass.** We now establish when the total mass of subordinate Brownian loop measure is finite. Recall that $\delta = 1$ when X has finite area and $\delta < 1$ when X has infinite area. The prime geodesic theorem gives the asymptotic growth rate of the number of primitive closed geodesics,

$$N_X(R) := \#\{\gamma \in \mathcal{P}_X : \ell_\gamma \le R\} \sim \frac{e^{\delta R}}{\delta R} \qquad \text{as } R \to \infty, \tag{40}$$

with the number growing exponentially at rate δ. This is the analogue of the prime number theorem, with primitive closed geodesics playing the role of primes.

Corollary 4.7 (Finiteness)**.** *Let ϕ be any of the Bernstein functions treated in this paper, with associated spectral parameter $s = s(\phi)$ (i.e., $s = 1$ for Brownian motion and α-stable, $s = \frac{1}{2} + \sqrt{\frac{1}{4} + \kappa}$ for killing with rate $\kappa \ge -\frac{1}{4}$ and for the shifted α-stable case). If $s(\phi) > \delta$ then the total mass is finite,*

$$\sum_{\gamma \in \mathcal{P}_X} \sum_{m=1}^{\infty} \mu_X^{\phi}(\mathcal{C}_X(\gamma^m)) < \infty. \tag{41}$$

Proof. In each case

$$\mu_X^{\phi}(\mathcal{C}_X(\gamma^m)) = \frac{C}{m} \cdot \frac{e^{(1-s)L}}{e^L - 1}, \qquad L = m\ell_\gamma, \quad s = s(\phi),$$

for a constant $C > 0$, equal to 1 in the Brownian and killing cases and to $\alpha/2$ in the stable cases. *Step 1: Summing over the iterates* m. For $L \ge \ell_{\text{sys}}$, where ℓ_{sys} is the length of the shortest closed geodesic on X (i.e., the systole), we have $e^L - 1 \ge (1 - e^{-\ell_{\text{sys}}})e^L$, so $\frac{e^{(1-s)L}}{e^L - 1} \le \frac{e^{-sL}}{1 - e^{-\ell_{\text{sys}}}}$. Using $\sum_{m \ge 1} \frac{x^m}{m} = -\log(1 - x)$ where $x = e^{-s\ell_\gamma}$,

$$\sum_{m \ge 1} \mu_X^{\phi}(\mathcal{C}_X(\gamma^m)) \le \frac{C}{1 - e^{-\ell_{\text{sys}}}} \sum_{m \ge 1} \frac{e^{-sm\ell_\gamma}}{m} = -\frac{C}{1 - e^{-\ell_{\text{sys}}}} \log\bigl(1 - e^{-s\ell_\gamma}\bigr).$$

Conversely, keeping only the $m = 1$ term gives

$$\sum_{m \ge 1} \mu_X^{\phi}(\mathcal{C}_X(\gamma^m)) \ge C \cdot \frac{e^{(1-s)\ell_\gamma}}{e^{\ell_\gamma} - 1} \ge C\, e^{-s\ell_\gamma}.$$

Since $N_X(R) < \infty$ for every R, only finitely many geodesics have length below any given bound, so $\ell_\gamma \to \infty$ along $\mathcal{P}_X$; as $e^{-s\ell_\gamma} \to 0$ and $-\log(1 - x) = x + O(x^2)$, the upper bound is asymptotic to $\frac{C}{1 - e^{-\ell_{\text{sys}}}} e^{-s\ell_\gamma}$. The total mass is therefore finite if and only if

$$\sum_{\gamma \in \mathcal{P}_X} e^{-s\ell_\gamma} < \infty. \tag{42}$$

Step 2: The geodesic sum via the counting function. Writing the sum (42) as an integral against N_X and integrating by parts over $[0, T]$,

$$\sum_{\ell_\gamma \le T} e^{-s\ell_\gamma} = \int_0^T e^{-sR}\, dN_X(R) = e^{-sT} N_X(T) + s \int_0^T e^{-sR}\, N_X(R)\, dR,$$

since $N_X(R) = 0$ for $R < \ell_{sys}$. By the prime geodesic theorem (40), $N_X(R) \asymp e^{\delta R}/R$ for large R, so the large-R behaviour of the integrand is $e^{-(s-\delta)R}/R$, and

$$\int^{\infty} \frac{e^{-(s-\delta)R}}{R}\, dR \quad \begin{cases} \text{converges}, & s > \delta, \\ \text{diverges (like } \int^{\infty} dR/R), & s = \delta, \\ \text{diverges}, & s < \delta. \end{cases}$$

For $s > \delta$ the boundary term $e^{-sT} N_X(T)$ tends to 0 as $T \to \infty$, so (42) converges; for $s \leq \delta$ the integral term alone already diverges as $T \to \infty$. Thus the total mass is finite when the decay rate s exceeds the proliferation rate δ of closed geodesics. For $s > \delta$ this gives (41). At $s = \delta$ the sum diverges; since $-\log Z_X(s)$ increases to this divergent sum as $s \downarrow \delta$, by monotone convergence $Z_X(s) \to 0$ as $s \downarrow \delta$, and the total mass blows up. In the finite-area case $\delta = 1$ and a killing rate $\kappa > 0$, equivalently $s(\kappa) > 1$, is needed to restore finiteness. □

5. Renormalising the Total Mass of Brownian Loops

5.1. **Zeta-regularised determinant of the Laplacian in the compact case.** In this section we consider closed (i.e., compact and without boundary) hyperbolic surfaces $X = \Gamma\backslash\mathbb{H}^2$ and express their zeta-regularised determinants of the Laplacian in terms of the total mass of Brownian loops.

Recall that on a compact hyperbolic surface the spectrum of Δ_X is an increasing discrete sequence of eigenvalues, counted with multiplicity,

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots.$$

The zero eigenvalue is simple, corresponding to constant eigenfunctions, and Weyl's law gives the rate at which the rest of the eigenvalues grow, $\lambda_j \sim 4\pi j/\operatorname{Area}(X)$ as $j \to \infty$.

An important quantity in spectral geometry and theoretical physics is the determinant of the Laplacian, $\det \Delta_X$, which one naively defines as the product of non-zero eigenvalues; taking the logarithm, we see that $\log\det\Delta_X \overset{!}{=} \sum_{j\geq 1} \log\lambda_j$ diverges. Ray and Singer [**?**], in their work on analytic torsion, instead defined the determinant through the spectral zeta function

$$\zeta_X(s) := \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s}, \qquad \operatorname{Re}(s) > 1,$$

using the identity $-\zeta_X'(0) = \sum_{j\geq 1} \log\lambda_j$ to set

$$\log\det\nolimits_\zeta \Delta_X := -\zeta_X'(0).$$

By the small time asymptotics of the heat trace we can see that ζ_X has a meromorphic continuation to $\mathbb{C}$. More specifically, via the Mellin transform we can write each λ_j^{-s} as $\Gamma(s)^{-1}\int_0^\infty t^{s-1}e^{-t\lambda_j}\, dt$ and sum over $j \geq 1$,

$$\zeta_X(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \sum_{j=1}^{\infty} e^{-t\lambda_j}\, dt = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1}\big(\operatorname{Tr}(e^{-t\Delta_X}) - 1\big)\, dt,$$

where we subtract 1 (i.e., $\dim\ker\Delta_X$) to drop the zero eigenvalue λ_0. As $t \downarrow 0$ the heat trace behaves like

$$\operatorname{Tr}(e^{-t\Delta_X}) - 1 \sim \frac{\operatorname{Area}(X)}{4\pi t} + \left(\frac{\chi(X)}{6} - 1\right) + O(t),$$

which determines the behaviour of the integral near $t = 0$. The t^{-1} term produces a simple pole of ζ_X at $s = 1$, and we can meromorphically continue ζ_X to $\mathbb{C}$. The constant term would give ζ_X a pole at $s = 0$, but the simple zero of $1/\Gamma(s)$ there cancels it. Hence ζ_X is analytic at $s = 0$, with $\zeta_X(0) = \chi(X)/6 - 1$, and $\zeta_X'(0)$ is well defined.

Since the work of Ray and Singer, zeta-regularised determinants have become widespread in theoretical physics, catalysed by Polyakov's paper on the quantum geometry of bosonic strings. For our case, it was observed in [**?**, LJ11] that $\log\det_\zeta \Delta$ can be interpreted as the total mass of the Brownian loop measure; a method provided in [**?**] for Riemann surfaces gives one form of renormalisation of the total mass, with the truncation done via quadratic variation. We will perform a different truncation, following [WX25], according to the length spectrum of the hyperbolic surface X. One of the motivations for this section is that, as per Theorem 4.7, the total mass of Brownian loop measure is infinite for finite-area surfaces. Expressing the total mass in terms of the zeta-regularised determinant provides a way to renormalise this mass, thus allowing one to define a well-defined probability measure on homotopy and homology classes.

As in [WX25], we use a refined prime geodesic theorem, which states that for a closed hyperbolic surface, where the critical exponent is $\delta = 1$,

$$N_X(R) \;:=\; \#\{\gamma \in \mathcal{P}_X : \ell_\gamma \le R\} \;=\; \mathrm{Li}(e^{\delta R}) + \sum_{0<\lambda_j\le 1/4} \mathrm{Li}(e^{s_j R}) + O_X\big(e^{3R/4}/R\big) \quad \text{as } R \to \infty, \tag{43}$$

where the logarithmic integral $\mathrm{Li}(x) = \int_2^x dt/\log t \sim x/\log x$ as $x \to \infty$ and $s_j = \frac{1}{2} + \sqrt{\frac{1}{4} - \lambda_j}$.

The Selberg trace formula for the heat semigroup in the compact case shows that

$$\begin{aligned}\sum_{j=0}^{\infty} e^{-t\lambda_j} \;&=\; \mathrm{Area}(X)\,\frac{e^{-t/4}}{(4\pi t)^{3/2}}\int_0^\infty \frac{r\,e^{-r^2/(4t)}}{\sinh(r/2)}\,dr \\ &+ \sum_{\gamma\in\mathcal{P}_X}\sum_{m=1}^{\infty} \frac{e^{-t/4}}{(4\pi t)^{1/2}}\,\frac{\ell(\gamma)}{2\sinh(m\ell(\gamma)/2)}\,e^{-(m\ell(\gamma))^2/(4t)},\end{aligned} \tag{44}$$

in which the first term on the right hand side is the identity contribution and the second term is the geometric contribution from hyperbolic conjugacy classes. Naud uses this to write $-\log\det_\zeta \Delta_X$ as a sum over the length spectrum,

$$-\log\det_\zeta \Delta_X \;=\; -\mathrm{Area}(X)\,E - \gamma_{EM} + \int_0^1 \frac{S_X(t)}{t}\,dt + \int_1^\infty \frac{S_X(t) - 1}{t}\,dt, \tag{45}$$

where $E = (4\zeta_R'(-1) - \frac{1}{2} + \log(2\pi))/(4\pi) \approx 0.0538$, $\gamma_{EM} \approx 0.5772$ is the Euler–Mascheroni constant, and

$$S_X(t) \;=\; \sum_{\gamma\in\mathcal{P}_X}\sum_{m=1}^{\infty} \frac{e^{-t/4}}{(4\pi t)^{1/2}}\,\frac{\ell(\gamma)}{2\sinh(m\ell(\gamma)/2)}\,e^{-(m\ell(\gamma))^2/(4t)}$$

is the geometric term of (44). Note that $S_X(t)$ is exponentially small as $t \to 0$, and $|S_X(t) - 1|$ is exponentially small as $t \to \infty$. With this we can re-write (45) in terms of subordinate Brownian loop measure.

Theorem 5.1 (Zeta-regularised determinant of the Laplacian in terms of subordinate Brownian loop measure, the compact case). *Let $X = \Gamma\backslash\mathbb{H}^2$ be a closed hyperbolic surface of genus g, and $\mathcal{G}(X)$ the set of all oriented closed geodesics on X. Write $\det_\zeta \Delta$ for the zeta-regularised determinant of Δ_X with the zero eigenvalue $\lambda_0 = 0$ excluded. Let ϕ be any of the Bernstein functions treated in this paper.*

(i) ***Brownian*** *($\phi(\lambda)=\lambda$):*

$$\begin{aligned} -\log \det_\zeta \Delta \;&=\; -\operatorname{Area}(X)\,E + C + \sum_{\gamma\in\mathcal{G}(X)\setminus\mathcal{P}_X} \mu_X(\mathcal{C}_X(\gamma)) \\ &\quad + \int_{R=0}^{\infty} \frac{1}{e^R-1}\, d\big(N_X(R) - \widetilde{\operatorname{Li}}(e^R)\big). \end{aligned} \tag{46}$$

(ii) ***Brownian with killing*** *($\phi(\lambda)=\lambda+\kappa$, $\kappa>0$). The determinant is recovered in the limit $\kappa\to 0^+$. For each $\kappa>0$,*

$$-\log \det_\zeta \Delta = -\operatorname{Area}(X)\,E + \log\kappa + \sum_{\gamma\in\mathcal{P}_X}\sum_{m=1}^{\infty} \mu_X^{\kappa}(\mathcal{C}_X(\gamma^m)) + O(\kappa) \tag{47}$$

$$= -\operatorname{Area}(X)\,E + \log\kappa - \log Z_X\big(\tfrac12 + \sqrt{\tfrac14+\kappa}\big) + O(\kappa), \tag{48}$$

and letting $\kappa\to 0^+$, where $s=\frac12+\sqrt{\frac14+\kappa}\to 1$ and the simple zero of Z_X at $s=1$ (from $\lambda_0=0$) gives $-\log Z_X(s)\sim -\log Z_X'(1) - \log\kappa$, the $\log\kappa$ terms cancel and the $O(\kappa)$ vanishes, leaving

$$\log \det_\zeta \Delta \;=\; \operatorname{Area}(X)\,E + \log Z_X'(1). \tag{49}$$

(iii) *α-**stable** ($\phi(\lambda)=\lambda^{\alpha/2}$, $\alpha\in(0,2)$):*

$$\begin{aligned} -\log \det_\zeta \Delta^{\alpha/2} \;&=\; \tfrac{\alpha}{2}\big(-\operatorname{Area}(X)\,E + C\big) + \sum_{\gamma\in\mathcal{G}(X)\setminus\mathcal{P}_X} \mu_X^{\alpha}(\mathcal{C}_X(\gamma)) \\ &\quad + \tfrac{\alpha}{2}\int_{R=0}^{\infty} \frac{1}{e^R-1}\, d\big(N_X(R) - \widetilde{\operatorname{Li}}(e^R)\big). \end{aligned} \tag{50}$$

Here $E=\frac{1}{4\pi}\big(4\zeta_R'(-1)-\frac12+\log(2\pi)\big)$ is the constant of (45), C is a universal constant, and $\widetilde{\operatorname{Li}}(x)$ is the cutoff of $\operatorname{Li}(x)$ at $x=2$,

$$\widetilde{\operatorname{Li}}(x) = \begin{cases} \int_2^x \frac{dt}{\log t} & \text{if } x\geq 2,\\ 0 & \text{if } x<2. \end{cases}$$

In (iii), $\det_\zeta \Delta^{\alpha/2}$ is the zeta-regularised determinant of the spectral fractional Laplacian, defined as in (45) with λ_j replaced by $\lambda_j^{\alpha/2}$. In (i) and (iii) the summation and integral converge.

Remark 5.2. Equation (49) is the classical determinant formula of D'Hoker–Phong [?]

$$\det_\zeta \Delta = Z_X'(1)\, e^{(2g-2)\left(2\zeta_R'(-1)-\frac14+\frac12\log 2\pi\right)}.$$

Proof of (i). The integral $\int_{R=0}^{\infty}\frac{1}{e^R-1}\,dN_X(R)$ is the total mass $\sum_{\gamma\in\mathcal{P}_X}\mu_X(\mathcal{C}_X(\gamma))$ of Brownian loops homotopic to a primitive geodesic. Subtracting from it $\int_{R=0}^{\infty}\frac{1}{e^R-1}\,d\widetilde{\operatorname{Li}}(e^R)$ renormalises the contribution of Brownian loops that are homotopic to a long ($R\gg 1$) primitive geodesic as suggested by (43). By the prime geodesic theorem (43), $|N_X(R)-\widetilde{\operatorname{Li}}(e^R)| = O_X(e^{(1-\epsilon)R})$ as $R\to\infty$ for some $\epsilon>0$ depending on X. Hence the integral in (46) converges by integration by parts. The sum converges for the same reason.

We split the integrals in (45) as

$$\int_0^1 \frac{S_X(t)}{t}\,dt + \int_1^\infty \frac{S_X(t)-1}{t}\,dt = \int_0^\infty \frac{S_X(t)-S_X^p(t)}{t}\,dt + \int_0^1 \frac{S_X^p(t)}{t}\,dt + \int_1^\infty \frac{S_X^p(t)-1}{t}\,dt, \tag{51}$$

where

$$S_X^p(t) = \sum_{\gamma \in \mathcal{P}_X} \frac{e^{-t/4}}{(4\pi t)^{1/2}} \frac{\ell(\gamma)}{2\sinh(\ell(\gamma)/2)} e^{-\frac{\ell(\gamma)^2}{4t}}$$

is the primitive part of $S_X(t)$, and evaluate each piece. The non-primitive part converges without renormalisation

$$\begin{aligned}\int_0^\infty \frac{S_X(t) - S_X^p(t)}{t}\, dt &= \sum_{\gamma \in \mathcal{P}_X} \sum_{m=2}^\infty \int_0^\infty \frac{1}{t} \frac{e^{-t/4}}{(4\pi t)^{1/2}} \frac{\ell(\gamma)}{2\sinh(m\ell(\gamma)/2)} e^{-\frac{(m\ell(\gamma))^2}{4t}}\, dt \\ &= \sum_{\gamma \in \mathcal{P}_X} \sum_{m=2}^\infty \frac{1}{m} \frac{1}{e^{m\ell(\gamma)} - 1} = \sum_{\gamma \in \mathcal{G}(X) \setminus \mathcal{P}_X} \mu_X(\mathcal{C}_X(\gamma)).\end{aligned} \tag{52}$$

For the primitive part, write $S_X^p(t)$ as an integral against the prime geodesic counting measure,

$$S_X^p(t) = \int_{R=0}^\infty \frac{e^{-t/4}}{(4\pi t)^{1/2}} \frac{R}{2\sinh(R/2)} e^{-\frac{R^2}{4t}}\, dN_X(R),$$

and exchange the order of integration in each of $\int_0^1 S_X^p(t)/t\, dt$ and $\int_1^\infty (S_X^p(t) - 1)/t\, dt$. The inner t-integrals evaluate via the error function; the full calculation is carried out in [WX25, Eqs. (4.13)–(4.16)]. Decomposing $N_X(R) = \widetilde{\mathrm{Li}}(e^R) + (N_X(R) - \widetilde{\mathrm{Li}}(e^R))$, the $d\widetilde{\mathrm{Li}}(e^R)$ part has no X-dependence and contributes a universal constant C_1, and in the $d(N_X(R) - \widetilde{\mathrm{Li}}(e^R))$ part the error-function expression collapses to $1/(e^R - 1)$, with convergence given by (43). Hence

$$\int_0^1 \frac{S_X^p(t)}{t}\, dt + \int_1^\infty \frac{S_X^p(t) - 1}{t}\, dt = C_1 + \int_{R=0}^\infty \frac{1}{e^R - 1}\, d\big(N_X(R) - \widetilde{\mathrm{Li}}(e^R)\big). \tag{53}$$

Substituting (52) and (53) into (51) and combining with (45) gives (46) with $C = -\gamma_{EM} + C_1$. See [WX25] for the exact constants. □

Proof of (ii). For $\kappa > 0$ the total mass is finite, so no cutoff is needed. Since the Brownian with killing heat trace is $e^{-\kappa t} S_X(t)$,

$$M_\kappa := \sum_{\gamma \in \mathcal{P}_X} \sum_{m=1}^\infty \mu_X^\kappa(\mathcal{C}_X(\gamma^m)) = \int_0^\infty e^{-\kappa t} \frac{S_X(t)}{t}\, dt.$$

Splitting at $t = 1$ and subtracting 1 from S_X in the tail,

$$M_\kappa = \int_0^1 e^{-\kappa t} \frac{S_X(t)}{t}\, dt + \int_1^\infty e^{-\kappa t} \frac{S_X(t) - 1}{t}\, dt + E_1(\kappa), \tag{54}$$

where $E_1(\kappa) := \int_1^\infty e^{-\kappa t}/t\, dt$ is the exponential integral. Comparing with the integral expression in Naud's formula (45),

$$\int_0^1 \frac{S_X(t)}{t}\, dt + \int_1^\infty \frac{S_X(t) - 1}{t}\, dt = M_\kappa - E_1(\kappa) + R_\kappa, \tag{55}$$

where the correction is

$$R_\kappa := \int_0^1 (1 - e^{-\kappa t}) \frac{S_X(t)}{t}\, dt + \int_1^\infty (1 - e^{-\kappa t}) \frac{S_X(t) - 1}{t}\, dt.$$

Using $1 - e^{-\kappa t} \le \kappa t$ on $(0, \infty)$,

$$|R_\kappa| \le \kappa \int_0^1 S_X(t)\, dt + \kappa \int_1^\infty |S_X(t) - 1|\, dt = O(\kappa),$$

since $S_X(t)$ is exponentially small as $t \to 0^+$ and $|S_X(t) - 1|$ is exponentially small as $t \to \infty$, so both integrals are finite. The standard expansion gives $E_1(\kappa) = -\gamma_{EM} - \log\kappa + O(\kappa)$ as $\kappa \to 0^+$. Substituting (55) into (45),

$$-\log\det_\zeta \Delta \;=\; -\operatorname{Area}(X)\,E - \gamma_{EM} + M_\kappa - E_1(\kappa) + R_\kappa \;=\; -\operatorname{Area}(X)\,E + \log\kappa + M_\kappa + O(\kappa),$$

where the two γ_{EM} terms cancel. □

Proof of (iii). Since $\zeta_{\Delta^{\alpha/2}}(s) = \zeta_X(\alpha s/2)$, one has $\log\det_\zeta \Delta^{\alpha/2} = (\alpha/2)\log\det_\zeta \Delta$. Multiplying (46) by $\alpha/2$ and using $\mu_X^\alpha = (\alpha/2)\mu_X$ on each homotopy-class term gives the result. □

5.1.1. *Polyakov's conformal anomaly formula.* Theorem 5.1 computes $\log\det_\zeta \Delta$ on the hyperbolic representative of each conformal class on a closed surface of genus g, and Polyakov's formula gives the transformation law for $\log\det_\zeta \Delta$ under metric rescaling within a fixed conformal class.

Theorem 5.3 (Polyakov's conformal anomaly formula [?]; see also [?]). *Let g_0 and $g = e^{2\sigma} g_0$ be conformally equivalent smooth metrics on a closed surface X, with K_0 the Gauss curvature of g_0. Then*

$$\log\det_\zeta \Delta_X \;=\; -\frac{1}{12\pi}\int_X |\nabla_{g_0}\sigma|^2 \, d\,\mathrm{vol}_{g_0} \;-\; \frac{1}{6\pi}\int_X K_0\,\sigma \, d\,\mathrm{vol}_{g_0} \;+\; \log\frac{\mathrm{vol}_g(X)}{\mathrm{vol}_{g_0}(X)} \;+\; \log\det_\zeta \Delta_{g_0}. \tag{56}$$

Specialising to $g_0 = g_{hyp}$, the unique hyperbolic representative in the conformal class of X (with $K_0 \equiv -1$ and $\mathrm{vol}_{g_0}(X) = \operatorname{Area}(X) = 4\pi(g-1)$ by Gauss–Bonnet), the curvature coupling reduces to $\frac{1}{6\pi}\int_X \sigma\, dA_{hyp}$. Writing

$$\mathcal{P}_X(\sigma) \;:=\; -\frac{1}{12\pi}\int_X |\nabla\sigma|^2 \, dA_{hyp} \;+\; \frac{1}{6\pi}\int_X \sigma \, dA_{hyp} \;+\; \log\frac{\mathrm{vol}_g(X)}{4\pi(g-1)}$$

for the Polyakov correction relative to g_{hyp}, so that $\log\det_\zeta \Delta_g = \mathcal{P}_X(\sigma) + \log\det_\zeta \Delta_{g_{hyp}}$, Theorem 5.1 now extends to arbitrary metrics in the conformal class.

Corollary 5.4 (Polyakov's formula via Brownian loop measure). *Let X be a closed hyperbolic surface of genus g, and let $g = e^{2\sigma} g_{hyp}$ be any smooth metric in the conformal class of X. Then*

$$\begin{aligned}\log\det_\zeta \Delta_X \;=\; &\mathcal{P}_X(\sigma) \;+\; \operatorname{Area}(X)\,E \;-\; C \;-\; \sum_{\gamma\in\mathcal{G}(X)\setminus\mathcal{P}_X} \mu_X(\mathcal{C}_X(\gamma)) \\ &- \int_{R=0}^{\infty} \frac{1}{e^R - 1}\, d\big(N_X(R) - \widetilde{\mathrm{Li}}(e^R)\big),\end{aligned} \tag{57}$$

Equivalently, via the $\kappa \to 0^+$ limit in (49),

$$\log\det_\zeta \Delta_X \;=\; \mathcal{P}_X(\sigma) \;+\; \operatorname{Area}(X)\,E \;+\; \log Z_X'(1). \tag{58}$$

5.2. **Zeta-regularised determinant of the Laplacian in the non-compact case.** We now turn to non-compact hyperbolic surfaces of finite area, where the above construction breaks down as the Laplacian no longer has purely discrete spectrum. Alongside the L^2 eigenvalues starting at $\lambda_0 = 0$, there is now a continuous spectrum filling $[\frac{1}{4}, \infty)$, with multiplicity equal to the number of cusps. The generalised eigenfunctions of this continuous spectrum are the Eisenstein series $E_j(z,s)$, one per cusp, which solve $\Delta_X E_j = s(1-s)E_j$ but do not lie in L^2. Thus there is no longer a discrete sequence of eigenvalues to feed into $\sum_j \lambda_j^{-s}$ and the heat semigroup $e^{-t\Delta_X}$ is no longer trace class. As such, the determinant of the Laplacian must be constructed in a different way.

There are several routes we can take. One method is to define a relative determinant, comparing Δ_X to a model operator along the ends (cusps in finite-area case or funnels in infinite-area) so that the divergent parts cancel. We will instead follow the microlocal analytic method of Melrose, which keeps Δ_X itself and regularises the trace. The main insight is that a hyperbolic surface has a natural compactification $\bar{X}$ obtained by capping each end with a circle at infinity, and on $\bar{X}$ the heat kernel has a controlled asymptotic expansion at the ends. One integrates the diagonal of the kernel in a renormalised sense that removes the divergent part of that expansion. The explicit coordinates near the cusps and funnels can be found in [**?**].

The renormalised integral. Fix the compactification $\bar{X}$ with a boundary defining function x, which is a smooth function vanishing to first order at the ends, and a smooth density μ. For f with a controlled expansion at the ends, the integral $\int_X x^z f\mu$ converges when $\mathrm{Re}(z)$ is large and continues meromorphically in z, and its finite part at $z = 0$ is the renormalised integral

$$ {}^0\!\!\int_X f\mu := \operatorname*{FP}_{z=0} \int_X x^z f\mu. \tag{59} $$

This is the Riesz renormalisation; the Hadamard version cuts the ends off at $x \geq \epsilon$ and takes the finite part as $\epsilon \to 0$, and the two agree for the functions we use. Applied to the volume form it gives the renormalised area ${}^0\mathrm{Area}(g) := {}^0\!\!\int_X \mathrm{d}\,\mathrm{vol}_g$, which for a hyperbolic metric equals $-2\pi\chi(X)$ by the Gauss–Bonnet theorem for these metrics [**?**].

The renormalised trace and determinant. For a trace-class operator with continuous kernel, Lidskii's theorem computes the trace as the integral of the diagonal. On a cusped surface the heat kernel is not trace class, but its diagonal $p_X(t, z, z)$ still has a controlled expansion at the ends, so the divergent integral of the diagonal can be replaced by its renormalised value. This is the renormalised trace, or 0-trace,

$$ {}^0\mathrm{Tr}(e^{-t\Delta_X}) := {}^0\!\!\int_X p_X(t, z, z)\ \mathrm{d}\,\mathrm{vol}_g(z), \tag{60} $$

defined for each $t > 0$. As $t \to \infty$ it converges exponentially to the rank of the L^2 null space, and as $t \to 0$ it has an asymptotic expansion in powers of t and $t \log t$, the logarithmic terms coming from the cusps. Via the Mellin transform we get the zeta function

$$ \zeta_X^0(s) := \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1}\ {}^0\mathrm{Tr}\big(e^{-t\Delta_X} - \mathcal{P}\big)\ \mathrm{d}t, \tag{61} $$

where $\mathcal{P}$ is the projection onto the L^2 null space, subtracted so the $t \to \infty$ end converges. The short-time expansion gives ζ_X^0 a meromorphic continuation to $\mathbb{C}$ that is regular at the origin, and the renormalised determinant is

$$ \det\nolimits_0 \Delta_X := e^{-(\zeta_X^0)'(0)}. \tag{62} $$

On a closed surface the 0-trace is the ordinary trace and $\det_0$ reduces to the zeta-regularised determinant of the compact case.

5.2.1. *Relation with the Selberg zeta function.* Borthwick, Judge and Perry [**?**] relate $\det_0 \Delta_X$ to the Selberg zeta function Z_X through the resolvent $R_X(s) = (\Delta_X - s(1-s))^{-1}$. The determinant satisfies

$$ \left(\frac{1}{2s-1}\frac{\partial}{\partial s}\right)^2 \log \det\nolimits_0(\Delta_X - s(1-s)) = -\,{}^0\mathrm{Tr}\big(R_X(s)^2\big), \tag{63} $$

and integrating this in s fixes the determinant up to a factor $e^{M+Fs(1-s)}$ with two constants, giving the following.

Theorem 5.5 ([?, ?]). *Let X be a geometrically finite hyperbolic surface with n_C cusps and Euler characteristic $\chi = \chi(X)$. Then*

$$\det_0(\Delta_X - s(1-s)) = Z_X(s)\, e^{M+Fs(1-s)}\, G_\infty(s)^\chi \left(2^s \sqrt{\pi(s-\tfrac{1}{2})}\, \Gamma(s-\tfrac{1}{2})\right)^{-n_C}, \tag{64}$$

where $G_\infty(s) = (2\pi)^{-s}\Gamma(s)G(s)^2$ is built from the Barnes G-function, and

$$M = \chi(X)\left(\tfrac{1}{2}\log 2\pi - 2\zeta_R'(-1) + \tfrac{1}{4}\right), \qquad F = -\chi(X),$$

with ζ_R the Riemann zeta function. Consequently

$$\det_0 \Delta_X = \begin{cases} C_X\, Z_X'(1) & \text{if } \operatorname{Area}(X) < \infty, \\ C_X\, Z_X(1) & \text{if } \operatorname{Area}(X) = \infty, \end{cases} \qquad C_X = e^M\, (2\pi)^{-\chi(X)} \left(\sqrt{2}\,\pi\right)^{-n_C}.$$

Remark 5.6. The derivative $Z_X'(1)$ in the finite-area case appears because 0 lies in the spectrum of Δ_X when $\operatorname{Area}(X) < \infty$. There $\lambda_0 = 0$ gives Z_X a simple zero at $s = 1$, which is divided out to form $\det_0 \Delta_X$. In infinite area 0 is not an L^2 eigenvalue, $Z_X(1) \neq 0$, and no derivative is needed.

Taking $-\log$ of (64) and separating the $\log Z_X$ term gives

$$-\log\det_0(\Delta_X - s(1-s)) = -Fs(1-s) - M - \log Z_X(s) - D_X(s), \tag{65}$$

where

$$D_X(s) := \chi \log G_\infty(s) - \log\left(2^{sn_C}\left(\pi(s-\tfrac{1}{2})\right)^{n_C/2}\Gamma(s-\tfrac{1}{2})^{n_C}\right). \tag{66}$$

Theorem 5.7 (Zeta-regularised determinant via Brownian loop measure, finite-area case). *Let X be a geometrically finite hyperbolic surface of finite area, with n_C cusps and Euler characteristic $\chi = \chi(X)$, and let $M, F, D_X(s)$ be as in Theorem 5.5. For $\kappa \geq 0$, write $s = \frac{1}{2} + \sqrt{\frac{1}{4} + \kappa} > 1$, so that $s(s-1) = \kappa$ and $\Delta_X - s(1-s) = \Delta_X + \kappa$. Then*

$$-\log\det_0(\Delta_X + \kappa) = F\kappa - M + \sum_{\gamma \in \mathcal{P}_X} \sum_{m=1}^{\infty} \mu_X^\kappa(\mathcal{C}_X(\gamma^m)) - D_X(s). \tag{67}$$

Since 0 lies in the spectrum of Δ_X, the determinant of Δ_X itself is recovered by dividing out the simple zero,

$$\det_0 \Delta_X := \lim_{s\to 1} \frac{\det_0(\Delta_X - s(1-s))}{s(s-1)}.$$

As $\kappa \to 0^+$ this gives

$$\log\det_0 \Delta_X = M + D_X(1) + \log Z_X'(1) = \log C_X + \log Z_X'(1), \tag{68}$$

where $D_X(1) = -\chi\log(2\pi) - n_C \log\left(\sqrt{2}\,\pi\right)$ and C_X is as in Theorem 5.5.

Proof. Substituting $-\log Z_X(s) = \sum_{\gamma\in\mathcal{P}_X}\sum_{m=1}^{\infty} \mu_X^\kappa(\mathcal{C}_X(\gamma^m))$ into (65) and using $s(1-s) = -\kappa$ gives (67). For the limit, $\operatorname{Area}(X) < \infty$ forces a simple zero $Z_X(s) = Z_X'(1)(s-1) + O\left((s-1)^2\right)$, so

$$-\log Z_X(s) = -\log Z_X'(1) - \log(s-1) + O(s-1).$$

Dividing by $s(s-1)$ subtracts $\log\left(s(s-1)\right)$ from $\log\det_0(\Delta_X - s(1-s))$; since $s(s-1) = \kappa$ and $s - 1 \sim \kappa$ as $\kappa \to 0^+$, the resulting $-\log(s-1)$ cancels the divergence in $-\log Z_X(s)$. Together with $F\kappa \to 0$, $D_X(s) \to D_X(1)$, and $D_X(1) = \log C_X - M$, this yields (68). ∎

Remark 5.8 (The infinite-area case). When $\mathrm{Area}(X) = \infty$ one has $\delta < 1$, so by Theorem 4.7 the total mass of subordinate Brownian loop measure is already finite. The determinant identity then holds directly at $s = 1$ since 0 is not an L^2 eigenvalue and $Z_X(1) \neq 0$. The corresponding $-\log\det_0 \Delta_X$ via the loop mass and the resonance divisor of Z_X is given in Lemonde–Wang [LW26]. It can be easily recovered via Theorem 5.5. Moreover, the Polyakov conformal anomaly formula for non-compact surfaces can be found in [?].

6. A probability measure on homotopy and homology classes

6.1. **A probability measure on free homotopy classes.** With all the finiteness conditions established, we can now normalise the mass of Brownian loop measure to form a probability measure on free homotopy classes. The resulting weights, one mass per homotopy class, are natural in that their normalising constant is $-\log Z_X(s)$ of Corollary 4.3 and the moments of random variables are governed by the Selberg zeta function and its derivatives. The motivation for defining this probability measure is that it allows, via an explicit weighting per class, to better understand the geometry of the surface, such as the probability of intersections of closed geodesics. We define the probability measure as

$$\mathbb{P}_s\big(\mathcal{C}_X(\gamma^m)\big) := \frac{\mu_X^\kappa\big(\mathcal{C}_X(\gamma^m)\big)}{-\log Z_X(s)} = \frac{\mu_X^\kappa\big(\mathcal{C}_X(\gamma^m)\big)}{\sum_{\gamma\in\mathcal{P}_X}\sum_{m=1}^\infty \mu_X^\kappa\big(\mathcal{C}_X(\gamma^m)\big)}, \qquad s = \tfrac{1}{2} + \sqrt{\tfrac{1}{4}+\kappa},$$

where throughout this section $\kappa > 0$. Note that you can also do the $\kappa = 0$ case if you use the expressions from Section 5.

Under $\mathbb{P}_s$ each homotopy class $\mathcal{C}_X(\gamma^m)$ is assigned a probability proportional to its mass under the loop measure, and any function of the class becomes a random variable. The most natural one is the length $L := m\ell_\gamma$ of the geodesic representative, and we can easily compute its expectation, variance, and any moment.

The mass $\mu_X^\kappa\big(\mathcal{C}_X(\gamma^m)\big)$ depends on s via $e^{(1-s)m\ell_\gamma}$ and differentiating gives

$$\frac{d}{ds}\,\mu_X^\kappa\big(\mathcal{C}_X(\gamma^m)\big) = -(m\ell_\gamma)\,\mu_X^\kappa\big(\mathcal{C}_X(\gamma^m)\big). \tag{69}$$

Rather than treat each moment separately, observe that shifting the spectral parameter is the same as tilting by the length. Indeed, for every $r > 1 - s$,

$$\mathbb{E}_s\big[e^{-rL}\big] = \frac{\displaystyle\sum_{\gamma,m}\mu_X^\kappa\big(\mathcal{C}_X(\gamma^m)\big)\,e^{-r\,m\ell_\gamma}}{-\log Z_X(s)} = \frac{-\log Z_X(s+r)}{-\log Z_X(s)} = \frac{\log Z_X(s+r)}{\log Z_X(s)}, \tag{70}$$

because the summand at parameter s multiplied by $e^{-r\,m\ell_\gamma}$ is exactly the summand at parameter $s+r$. Writing $F(s) := -\log Z_X(s)$ for the total mass, (69) gives $\sum_{\gamma,m}\mu_X^\kappa\big(\mathcal{C}_X(\gamma^m)\big)(m\ell_\gamma)^n = (-1)^n F^{(n)}(s)$ and hence all moments at once,

$$\mathbb{E}_s\big[L^n\big] = \frac{(-1)^n F^{(n)}(s)}{F(s)}, \qquad n \geq 1. \tag{71}$$

The first two cumulants are then derivatives of $\log F$. For the mean,

$$\mathbb{E}_s[L] = -\frac{d}{ds}\log\big(-\log Z_X(s)\big) = -\frac{F'(s)}{F(s)} = -\frac{Z_X'(s)}{Z_X(s)\,\log Z_X(s)}, \tag{72}$$

and for the variance,

$$\mathrm{Var}_s(L) = \frac{d^2}{ds^2}\log\big(-\log Z_X(s)\big) = \frac{F''(s)F(s) - F'(s)^2}{F(s)^2} = \frac{\big(Z_X Z_X'' - (Z_X')^2\big)\log Z_X - (Z_X')^2}{Z_X^2\,\log^2 Z_X}. \quad (73)$$

In particular $\log F$ is strictly convex on $(1,\infty)$, so $s \mapsto \mathbb{E}_s[L]$ is strictly decreasing whereby increasing the killing rate shortens the typical class, as expected.

We can also see the effect of sending the killing rate to ∞. As $s \to \infty$ the weights $\mu_X^\kappa(\mathcal{C}_X(\gamma^m)) \sim e^{-s\,m\ell_\gamma}$ are dominated by the primitive classes ($m = 1$) whose geodesic realises the systole $\ell_{\mathrm{sys}} := \min_{\gamma\in\mathcal{P}_X}\ell_\gamma$. Since $\mathcal{P}_X$ consists of oriented primitive closed geodesics, and since a hyperbolic element of a torsion-free Fuchsian group is never conjugate to its inverse, the systolic length is attained by at least two classes. Writing

$$N_{\mathrm{sys}} := \#\big\{\gamma \in \mathcal{P}_X :\ \ell_\gamma = \ell_{\mathrm{sys}}\big\}\ \geq 2,$$

the measure $\mathbb{P}_s$ concentrates on the set of systolic classes and distributes itself among them,

$$\mathbb{P}_s\big(\mathcal{C}_X(\gamma)\big) \xrightarrow{s\to\infty} \frac{1}{N_{\mathrm{sys}}} \quad \text{for each } \gamma\in\mathcal{P}_X \text{ with } \ell_\gamma = \ell_{\mathrm{sys}}, \qquad \mathbb{P}_s\big(\mathcal{C}_X(\gamma^m)\big) \xrightarrow{s\to\infty} 0 \quad \text{for every other class.}$$

Consequently the mean length converges to the systole length,

$$\mathbb{E}_s[L] \xrightarrow{s\to\infty} \ell_{\mathrm{sys}}.$$

The same appears on the analytic side. As $s \to \infty$, the dominant contribution to

$$-\log Z_X(s) = \sum_{\gamma\in\mathcal{P}_X}\sum_{m\geq 1}\frac{1}{m}\,\frac{e^{(1-s)m\ell_\gamma}}{e^{m\ell_\gamma}-1}$$

comes from the N_{sys} primitive systolic terms, each of order $e^{-s\,\ell_{\mathrm{sys}}}$, so

$$-\log Z_X(s)\ \sim\ C\,e^{-s\,\ell_{\mathrm{sys}}}, \qquad C = \frac{N_{\mathrm{sys}}}{1-e^{-\ell_{\mathrm{sys}}}}, \qquad (s\to\infty),$$

and both the systole and the multiplicity with which it occurs are recovered from the asymptotics,

$$\ell_{\mathrm{sys}} = -\lim_{s\to\infty}\frac{1}{s}\log\big(-\log Z_X(s)\big), \qquad N_{\mathrm{sys}} = \big(1-e^{-\ell_{\mathrm{sys}}}\big)\lim_{s\to\infty} e^{s\ell_{\mathrm{sys}}}\big(-\log Z_X(s)\big).$$

6.2. **A probability measure on homology classes.** The homotopy-class decomposition of Section 3 is a very fine partition of the loop measure by topology, in which two loops contribute to the same term if and only if they are freely homotopic. More specifically, recall that for a surface X of genus $g \geq 2$ the fundamental group $\pi_1(X) \simeq \Gamma$ is non-abelian. A free homotopy class corresponds to a conjugacy class in Γ, and retains non-abelian information such as the order in which different handles are traversed, up to conjugation. Passing from free homotopy classes to homology classes discards this information and gives a much coarser equivalence relation.

The first homology group $H_1(X,\mathbb{Z})$ is the abelianisation of $\pi_1(X)$, so it records only the net winding around each cycle. For a closed surface of genus g one has $H_1(X,\mathbb{Z}) \simeq \mathbb{Z}^{2g}$, while for a geometrically finite non-compact surface with $b \geq 1$ ends, of which n_C are cusps and n_F are funnels, it has rank $2g + b - 1$. Since $b = n_C + n_F$, this gives $H_1(X,\mathbb{Z}) \simeq \mathbb{Z}^{2g+n_C+n_F-1}$. Via the Hurewicz theorem we write $\Gamma \twoheadrightarrow H_1(X,\mathbb{Z})$ for the abelianisation map and $[\gamma]$ for the image in homology of an oriented primitive closed geodesic $\gamma \in \mathcal{P}_X$. In particular, $[\gamma^m] = m[\gamma]$. Grouping the loop measure by homology class amounts to summing the homotopy-class contributions $\mu_X^\kappa(\mathcal{C}_X(\gamma^m))$ over all pairs (γ, m) for which $[\gamma^m]$ is equal to a fixed $\beta \in H_1(X,\mathbb{Z})$. The fixed homology class β has contributions from infinitely many distinct free homotopy classes. On the analytic side, the distribution of closed geodesics across homology classes can be detected using Selberg L-functions, which play the same role as Dirichlet L-functions do for primes in arithmetic progressions.

A character of $H_1(X,\mathbb{Z})$ is a homomorphism $\chi : H_1(X,\mathbb{Z}) \to \mathbb{C}^\times$, and the unitary characters $\chi : H_1(X,\mathbb{Z}) \to S^1$ form the character torus (i.e., the Pontryagin dual) $\widehat{H_1(X,\mathbb{Z})}$. Writing $r = 2g + n_C + n_F - 1$ for the rank when X has at least one end, a unitary character is determined by its values on a $\mathbb{Z}$-basis. More explicitly, after choosing a $\mathbb{Z}$-basis $e_1,\dots,e_r$ and phases $\theta_1,\dots,\theta_r \in \mathbb{R}/\mathbb{Z}$, one has $\chi(e_j) = e^{2\pi i\theta_j}$, so

$$\widehat{H_1(X,\mathbb{Z})} \simeq (S^1)^r \simeq (\mathbb{R}/\mathbb{Z})^r.$$

For a closed surface the rank is instead $r = 2g$, and the character torus has a remarkable relation to harmonic 1-forms and the Jacobian variety. By the Hodge theorem, every class in the de Rham cohomology group $H^1_{dR}(X,\mathbb{R})$ has a unique harmonic representative, so $H^1_{dR}(X,\mathbb{R}) \simeq \mathcal{H}^1(X)$, where $\mathcal{H}^1(X)$ denotes the space of real harmonic 1-forms. Under this identification, $H^1(X,\mathbb{Z})$ corresponds to the lattice $\mathcal{H}^1_{\mathbb{Z}}(X)$ of harmonic 1-forms with integral periods.

To a harmonic 1-form ω one associates the unitary character

$$\chi_\omega(\beta) = e^{2\pi i \int_\beta \omega}.$$

Here $\int_\beta \omega$ is the period of ω around the cycle β and the whole exponential is the unitary holonomy. Two harmonic 1-forms determine the same character exactly when their difference has integral periods. It follows that there are natural isomorphisms of compact real tori

$$\widehat{H_1(X,\mathbb{Z})} \simeq \frac{H^1_{dR}(X,\mathbb{R})}{H^1(X,\mathbb{Z})} \simeq \frac{\mathcal{H}^1(X)}{\mathcal{H}^1_{\mathbb{Z}}(X)} \simeq \mathrm{Jac}(X),$$

where $H^1(X,\mathbb{Z})$ is viewed inside $H^1_{dR}(X,\mathbb{R})$ via the de Rham isomorphism. The Hodge star satisfies $*^2 = -1$ on 1-forms and therefore endows the real torus of dimension $2g$ with the structure of a complex torus of dimension g. Together with the intersection pairing, this gives the Jacobian its structure as a principally polarised abelian variety.

When X is closed, under $\widehat{H_1(X,\mathbb{Z})} \simeq \mathrm{Jac}(X)$, the natural pairing is $\langle \beta, [\omega]\rangle = \int_\beta \omega \pmod{\mathbb{Z}}$, and the Fourier inversion formula (78) can therefore be written as an integral over $\mathrm{Jac}(X)$ against $e^{-2\pi i\langle\beta,[\omega]\rangle}$. In the geometrically finite non-compact case the identification with the Jacobian is not available; while partial analogues exist, they require much more Hodge-theoretic machinery than we would like to invoke in this paper.

Definition 6.1 (Mass of Brownian loop measure in a homology class)**.** For $\beta \in H_1(X,\mathbb{Z})$ and $s = \frac{1}{2} + \sqrt{\frac{1}{4}+\kappa}$ with $\mathrm{Re}(s) > \delta$, the mass of the Brownian loop measure with killing in homology class β is

$$\mu^\kappa_X(\beta) := \sum_{\substack{\gamma\in\mathcal{P}_X,\ m\geq 1\\ m[\gamma]=\beta}} \mu^\kappa_X\big(\mathcal{C}_X(\gamma^m)\big) = \sum_{\substack{\gamma\in\mathcal{P}_X,\ m\geq 1\\ m[\gamma]=\beta}} \frac{1}{m}\cdot\frac{e^{(1-s)m\ell_\gamma}}{e^{m\ell_\gamma}-1}. \tag{74}$$

Remark 6.2. The first definition of Brownian loop measure in homology classes appeared in, based on our knowledge, [LJ11]. The definition in [LJ11] was initially not clear to us due to differing conventions and techniques, so we developed our definition independently and since found that the methods in this section, such as invoking the Selberg L-function, provide a dual approach to [LJ11], recovering their results in greater generality (e.g., extending to the non-compact case).

One of the motivations for introducing $\mu_X^\kappa(\beta)$ is to analyse intersections of closed geodesics; while geometric intersection numbers are well-defined on free homotopy classes, algebraic intersection numbers are defined on homology classes. Moreover, the weight $\chi([\gamma])^m$ in the logarithmic expansion of the Selberg L-function below, see (76), depends only on the homology class of the iterate γ^m, rather than on the particular geodesic representative. Whenever $m[\gamma] = \beta$,

$$\chi([\gamma])^m = \chi(m[\gamma]) = \chi(\beta),$$

so the double sum in (76) may be regrouped by homology class. We first define the Selberg L-function.

Definition 6.3 (Selberg L-function)**.** The *Selberg* L*-function* associated with a unitary character $\chi : H_1(X, \mathbb{Z}) \to S^1$ is

$$L_X(s, \chi) := \prod_{\gamma \in \mathcal{P}_X} \prod_{k=0}^{\infty} \left(1 - \chi([\gamma])\, e^{-(s+k)\ell_\gamma}\right), \qquad \mathrm{Re}(s) > \delta. \tag{75}$$

When χ is trivial, $L_X(s, \chi) = Z_X(s)$. Thus $L_X(s, \chi)$ is the twisted Selberg zeta function associated with a one-dimensional abelian representation and it admits a meromorphic continuation to $\mathbb{C}$.

The logarithm of $L_X(s, \chi)$ has the same expansion as in the untwisted case. For $\mathrm{Re}(s) > \delta$ the Euler product (75) converges absolutely, so one may take logarithms term by term and expand each factor using $-\log(1 - z) = \sum_{m=1}^{\infty} \frac{z^m}{m}$ when $|z| < 1$. Here $z = \chi([\gamma])e^{-(s+k)\ell_\gamma}$, and since χ is unitary, $|\chi([\gamma])| = 1$, so $|z| = e^{-(\mathrm{Re}(s)+k)\ell_\gamma} < 1$. Summing over k gives

$$\sum_{k=0}^{\infty} e^{-(s+k)m\ell_\gamma} = \frac{e^{-sm\ell_\gamma}}{1 - e^{-m\ell_\gamma}} = \frac{e^{(1-s)m\ell_\gamma}}{e^{m\ell_\gamma} - 1},$$

which gives the following identity.

Corollary 6.4 (Selberg L-function identity)**.** *Let $\chi : H_1(X, \mathbb{Z}) \to S^1$ be a unitary character, and let $s = \frac{1}{2} + \sqrt{\frac{1}{4} + \kappa}$ with $\mathrm{Re}(s) > \delta$. Then*

$$-\log L_X(s, \chi) = \sum_{\gamma \in \mathcal{P}_X} \sum_{m=1}^{\infty} \chi([\gamma])^m \, \mu_X^\kappa\big(\mathcal{C}_X(\gamma^m)\big) = \sum_{\gamma \in \mathcal{P}_X} \sum_{m=1}^{\infty} \frac{1}{m} \cdot \frac{\chi([\gamma])^m \, e^{(1-s)m\ell_\gamma}}{e^{m\ell_\gamma} - 1}. \tag{76}$$

Theorem 6.5 (Fourier expansion and inversion by homology class)**.** *Let $X = \Gamma \backslash \mathbb{H}^2$ be a geometrically finite hyperbolic surface with $H_1(X, \mathbb{Z}) \simeq \mathbb{Z}^r$, where $r = 2g + b - 1$ for a non-compact surface of genus g with $b \geq 1$ ends and $r = 2g$ when X is closed. Let $\kappa \geq -\frac{1}{4}$ with $s = \frac{1}{2} + \sqrt{\frac{1}{4} + \kappa}$ satisfying $\mathrm{Re}(s) > \delta$. Then the logarithm of the Selberg L-function admits the absolutely convergent Fourier expansion*

$$-\log L_X(s, \chi) = \sum_{\beta \in H_1(X, \mathbb{Z})} \chi(\beta) \mu_X^\kappa(\beta) \tag{77}$$

for every unitary character $\chi \in \widehat{H_1(X, \mathbb{Z})}$. Moreover, for each $\beta \in H_1(X, \mathbb{Z})$ one has the inversion formula

$$\mu_X^\kappa(\beta) = \int_{\widehat{H_1(X, \mathbb{Z})}} \big(-\log L_X(s, \chi)\big) \overline{\chi(\beta)} \, d\chi, \tag{78}$$

where $d\chi$ *is the normalised Haar measure on* $\widehat{H_1(X,\mathbb{Z})} \simeq (S^1)^r$.

Proof. Starting from (76), we group together all terms for which the iterate γ^m has the same homology class $\beta = m[\gamma]$, thus

$$-\log L_X(s,\chi) = \sum_{\beta \in H_1(X,\mathbb{Z})} \Bigg(\sum_{\substack{\gamma \in \mathcal{P}_X,\, m \geq 1 \\ m[\gamma]=\beta}} \mu_X^\kappa(\mathcal{C}_X(\gamma^m)) \Bigg) \chi(\beta).$$

To recover the coefficient of a fixed homology class β, multiply both sides of (77) by $\overline{\chi(\beta)}$ and integrate over the character torus. Since the Fourier series is absolutely convergent, the sum and the integral may be interchanged, giving

$$\int_{\widehat{H_1(X,\mathbb{Z})}} \big(-\log L_X(s,\chi)\big)\, \overline{\chi(\beta)}\, d\chi = \sum_{\beta' \in H_1(X,\mathbb{Z})} \mu_X^\kappa(\beta') \int_{\widehat{H_1(X,\mathbb{Z})}} \chi(\beta')\overline{\chi(\beta)}\, d\chi.$$

By orthogonality of characters on $\widehat{H_1(X,\mathbb{Z})}$, the inner integral is 1 if $\beta' = \beta$ and 0 otherwise. Therefore only the term $\beta' = \beta$ survives, giving (78). ☐

Remark 6.6. In the closed case, the inversion formula may equivalently be written as

$$\mu_X^\kappa(\beta) = \int_{\mathrm{Jac}(X)} \big(-\log L_X(s,\chi_{[\omega]})\big) e^{-2\pi i \int_\beta \omega}\, d[\omega], \tag{79}$$

where $d[\omega]$ is the normalised Haar measure on the underlying real Jacobian torus.

Proposition 6.7 (Distribution of the total homology of the loop soup). *Let* $s = \frac{1}{2} + \sqrt{\frac{1}{4}+\kappa}$ *with* $\mathrm{Re}(s) > \delta$, *let* $\mathcal{L}_\lambda$ *be the loop soup of intensity* $\lambda > 0$, *and let*

$$\beta^{(\lambda)} := \sum_{\eta \in \mathcal{L}_\lambda^*} [\eta] \;\in\; H_1(X,\mathbb{Z})$$

be the total homology of the loop soup, where $\mathcal{L}_\lambda^*$ *denotes the loops of* $\mathcal{L}_\lambda$ *that are non-contractible and not homotopic into a cusp. The sum is finite, since* $\#\mathcal{L}_\lambda^*$ *is Poisson with finite mean* $\lambda \sum_{\gamma,m} \mu_X^\kappa(\mathcal{C}_X(\gamma^m)) = -\lambda \log Z_X(s)$. *Then for every unitary character* $\chi \in \widehat{H_1(X,\mathbb{Z})}$,

$$\mathbb{E}\big(\chi(\beta^{(\lambda)})\big) = \Big(\frac{Z_X(s)}{L_X(s,\chi)}\Big)^\lambda, \tag{80}$$

and consequently, for each $\beta \in H_1(X,\mathbb{Z})$,

$$\mathbb{P}\big(\beta^{(\lambda)} = \beta\big) = Z_X(s)^\lambda \int_{\widehat{H_1(X,\mathbb{Z})}} L_X(s,\chi)^{-\lambda}\, \overline{\chi(\beta)}\, d\chi, \tag{81}$$

where $d\chi$ *is the normalised Haar measure on* $\widehat{H_1(X,\mathbb{Z})} \simeq (S^1)^r$, *and the complex powers are defined via* $L_X(s,\chi)^{-\lambda} := \exp\big(-\lambda \log L_X(s,\chi)\big)$ *with* $\log L_X(s,\chi)$ *given by* (76).

Proof. For a Poisson process of intensity $\lambda\mu_X^\kappa$ and any measurable F on loops, the exponential formula gives

$$\mathbb{E}\Big(\prod_{\eta \in \mathcal{L}_\lambda} e^{F(\eta)}\Big) = \exp\Big(\lambda \int \big(e^{F(\eta)} - 1\big)\, \mu_X^\kappa(d\eta)\Big).$$

Apply this with $e^{F(\eta)} = \chi([\eta])$ for $\eta \in \mathcal{L}_\lambda^*$, $\prod_\eta e^{F(\eta)} = \chi\big(\sum_{\eta \in \mathcal{L}_\lambda^*}[\eta]\big) = \chi(\beta^{(\lambda)})$, and the right-hand side is

$$\exp\Big(\lambda \sum_{\gamma \in \mathcal{P}_X} \sum_{m \geq 1} \big(\chi([\gamma])^m - 1\big)\, \mu_X^\kappa(\mathcal{C}_X(\gamma^m))\Big) = \exp\big(-\lambda \log L_X(s,\chi) + \lambda \log Z_X(s)\big) = \Big(\frac{Z_X(s)}{L_X(s,\chi)}\Big)^\lambda,$$

by the Selberg L-function identity applied to χ and to the trivial character. This proves (80). Multiplying by $\overline{\chi(\beta)}$ and integrating over $\widehat{H_1(X,\mathbb{Z})}$, orthogonality of characters isolates the class β and gives (81). $\square$

7. Brownian loops on hyperbolic 3-manifolds

The construction of the Brownian loop measure in Section 2.1 used the surface X only through its heat kernel, the bridge measures, the multiplicative Haar measure dt/t, and the Riemannian volume measure vol_g. But notice that the heat kernel exists on any complete Riemannian manifold, the bridge measures are the disintegrations of the path law with respect to the endpoint, and the two weights are measures on $(0,\infty)$ and on the manifold. The homotopy-class decomposition of Section 3 likewise is not confined to surfaces, relying on the descent (11) and the unfolding over cosets of a cyclic centraliser.

What tied us to surfaces was the conformal invariance of the Brownian loop measure; the Polyakov conformal anomaly formula and the proof of the length spectra identity of [WX25] relied on it, but once one works with a killing rate, or with any nonlinear subordination, conformal invariance breaks down and nothing ties our construction to surfaces.

So in this section we take $X = \Gamma\backslash\mathbb{H}^3$, where $\mathbb{H}^3$ is hyperbolic 3-space and $\Gamma \subset \mathrm{PSL}(2,\mathbb{C})$ is a torsion-free Kleinian group, so that X is a complete orientable hyperbolic 3-manifold. The loop measure and its subordinate versions are defined exactly as before. One difference from the surface case is that non-parabolic, non-elliptic elements of Γ are loxodromic; they translate along a geodesic axis and may also rotate about that axis. Thus an oriented closed geodesic carries a complex length

$$L_\gamma = \ell_\gamma + i\theta_\gamma, \qquad \theta_\gamma \in \mathbb{R}/2\pi\mathbb{Z},$$

where ℓ_γ is its translation length and θ_γ is its holonomy angle. For a geometrically finite hyperbolic 3-manifold, free homotopy classes of oriented closed curves correspond to conjugacy classes in Γ. The non-trivial, non-peripheral classes correspond to loxodromic conjugacy classes, and each such class contains a unique oriented closed geodesic representative.

7.1. **General homotopy class decomposition for hyperbolic 3-manifolds.** The setup is largely as in Section 3. We write $\mathcal{P}_X$ for the set of primitive oriented closed geodesics on X. Each $\gamma \in \mathcal{P}_X$ corresponds to a primitive loxodromic conjugacy class in Γ with complex length $L_\gamma = \ell_\gamma + i\theta_\gamma$, the real part ℓ_γ being the translation length and the imaginary part $\theta_\gamma \in \mathbb{R}/2\pi\mathbb{Z}$ the holonomy rotation about the geodesic. We use the upper half-space model $\mathbb{H}^3 = \{(z,y) : z \in \mathbb{C},\ y > 0\}$. Fixing a representative $\tau \in \Gamma$ and conjugating in $\mathrm{PSL}(2,\mathbb{C})$, we may place τ in the standard form

$$\tau : (z,y) \longmapsto \left(e^{L_\gamma} z,\ e^{\ell_\gamma} y\right), \tag{82}$$

the loxodromic isometry of $\mathbb{H}^3$ whose axis is the vertical geodesic from 0 to ∞, acting along it by translation through ℓ_γ and rotation through θ_γ. We again write $\mathcal{C}_X(\gamma^m)$ for the free non-peripheral homotopy class of closed curves winding m times around γ, corresponding to the conjugacy class of τ^m,

$$[\tau^m]_{\mathrm{conj}} := \{h\tau^m h^{-1} : h \in \Gamma\}.$$

Since Γ is torsion-free and discrete, anything commuting with τ^m preserves the axis of τ, and the elements of Γ preserving that axis form an infinite cyclic subgroup generated by the primitive element τ,

$$C_\Gamma(\tau^m) = \langle\tau\rangle = \{\tau^k : k \in \mathbb{Z}\}.$$

Enumerating the elements of $[\tau^m]_{\mathrm{conj}}$ without repetition, two elements $h_1, h_2 \in \Gamma$ produce the same conjugate, $h_1\tau^m h_1^{-1} = h_2\tau^m h_2^{-1}$, if and only if $h_1^{-1}h_2 \in \langle\tau\rangle$, that is if and only if h_1 and h_2 lie in the same left coset of $\langle\tau\rangle$ in Γ. Consequently

$$[\tau^m]_{\mathrm{conj}} = \bigsqcup_{r\in\Gamma/\langle\tau\rangle} \{r\tau^m r^{-1}\}, \tag{83}$$

with one distinct conjugate per coset.

We also assume the heat kernel $p^{\mathcal{E}}_{\mathbb{H}^3}$ decays fast enough in its spatial variables that, together with the discreteness of Γ, the periodisation converges absolutely.

The fundamental region. In the standard form (82), τ scales the height by e^{ℓ_γ} and rotates the horizontal coordinate by θ_γ. The height is scaled by the real factor e^{ℓ_γ}, so each orbit of $\langle\tau\rangle$ meets the slab $1 \le y < e^{\ell_\gamma}$ in exactly one point, and we take

$$F_\tau := \{(z, y) \in \mathbb{H}^3 : 1 \le y < e^{\ell_\gamma}\} \tag{84}$$

as a fundamental region for $\langle\tau\rangle$ acting on $\mathbb{H}^3$. Note that the rotation θ_γ acts within each slab and does not affect which slab a point lies in.

Theorem 7.1 (General homotopy class decomposition for hyperbolic 3-manifolds)**.** *Let $\gamma \in \mathcal{P}_X$ be a primitive closed geodesic with loxodromic representative $\tau \in \Gamma$, normalised as in (82), and let $m \ge 1$ be the winding number. The mass of the Dirichlet form loop measure in the free non-peripheral homotopy class $\mathcal{C}_X(\gamma^m)$ is*

$$\mu^{\mathcal{E}}_X(\mathcal{C}_X(\gamma^m)) = \int_0^\infty \frac{dt}{t} \int_{F_\tau} p^{\mathcal{E}}_{\mathbb{H}^3}(t, w, \tau^m w)\, d\,\mathrm{vol}_{\mathbb{H}^3}(w). \tag{85}$$

Proof. Identical in structure to the proof of Theorem 3.2, using the loxodromic standard form (82). ∎

7.2. **Subordinate Brownian cases for hyperbolic 3-manifolds.** We now specialise to subordinate Brownian loop measure. Applying Theorem 7.1 with $p^{\mathcal{E}}_{\mathbb{H}^3} = p^{\phi}_{\mathbb{H}^3}$ and expanding via (6) gives

$$\mu^{\phi}_X(\mathcal{C}_X(\gamma^m)) = \int_0^\infty \frac{dt}{t} \int_{F_\tau} \int_{[0,\infty)} p_{\mathbb{H}^3}(s, w, \tau^m w)\, \psi^{\phi}_t(\,ds)\, d\,\mathrm{vol}_{\mathbb{H}^3}(w). \tag{86}$$

As before, the strategy is to evaluate the spatial integral using an identity for the Brownian heat kernel on $\mathbb{H}^3$, then collapse the dt/t-integral into the weighted potential measure V_ϕ. We write $L := mL_\gamma = m\ell_\gamma + im\theta_\gamma$ for the complex length of the m-fold iterate. In the surface case we relied on the identity of [WX25], but on $\mathbb{H}^3$ we derive the corresponding identity ourselves.

Recall that the Brownian heat kernel on $\mathbb{H}^3$ is

$$p_{\mathbb{H}^3}(t, z, w) = \frac{1}{(4\pi t)^{3/2}} \frac{u}{\sinh u}\, e^{-t-\frac{u^2}{4t}}, \qquad u = d(z, w), \tag{87}$$

with $d(z, w)$ the hyperbolic distance. Work in the upper half-space model $\mathbb{H}^3 = \{(z, y) : z \in \mathbb{C},\ y > 0\}$ with τ in the standard form (82), so that $\tau^m(z, y) = (e^L z,\ e^{m\ell_\gamma} y)$. For $w = (z, y)$, the distance $u = d(w, \tau^m w)$ satisfies

$$\cosh u = 1 + \frac{|z - e^L z|^2 + (y - e^{m\ell_\gamma}y)^2}{2\,e^{m\ell_\gamma}y^2} = \cosh(m\ell_\gamma) + \frac{\left|e^L - 1\right|^2 |z|^2}{2\,e^{m\ell_\gamma}y^2},$$

using $\left|e^L - 1\right|^2 = 1 - 2e^{m\ell_\gamma}\cos(m\theta_\gamma) + e^{2m\ell_\gamma}$ and $1 + \frac{(1-e^{m\ell_\gamma})^2}{2e^{m\ell_\gamma}} = \cosh(m\ell_\gamma)$.

The fundamental region $F_\tau = \{(z, y) : 1 \le y < e^{\ell_\gamma}\}$ ranges over all $z \in \mathbb{C}$, and the volume element is $d\,\mathrm{vol}_{\mathbb{H}^3} = y^{-3}\, dA(z)\, dy$, where $dA(z)$ is Euclidean area measure on $\mathbb{C}$. In polar coordinates $z = re^{i\varphi}$, $dA(z) = r\, dr\, d\varphi$, and the distance u depends on z only through r, so the angular integral contributes 2π and

$$\int_{F_\tau} p_{\mathbb{H}^3}(t, w, \tau^m w)\, d\,\mathrm{vol}_{\mathbb{H}^3}(w) = 2\pi \int_1^{e^{\ell_\gamma}} \int_0^\infty p_{\mathbb{H}^3}(t, u)\, \frac{r\, dr\, dy}{y^3}, \qquad \cosh u = \cosh(m\ell_\gamma) + \frac{|e^L - 1|^2 r^2}{2\, e^{m\ell_\gamma} y^2}.$$

For fixed y we change variables from r to u. Differentiating, $\sinh u\, du = \dfrac{|e^L - 1|^2 r}{e^{m\ell_\gamma} y^2}\, dr$, so

$$r\, dr = \frac{e^{m\ell_\gamma} y^2}{|e^L - 1|^2} \sinh u\, du,$$

and as r runs from 0 to ∞, u runs from $m\ell_\gamma$ to ∞. The factor $\sinh u$ cancels the $1/\sinh u$ in (87), and the inner integral becomes

$$\int_0^\infty p_{\mathbb{H}^3}(t, u)\, r\, dr = \frac{e^{m\ell_\gamma} y^2}{|e^L - 1|^2} \frac{2t\, e^{-t}}{(4\pi t)^{3/2}}\, e^{-(m\ell_\gamma)^2/4t}.$$

The y^2 meets the y^{-3} of the volume element, and $\int_1^{e^{\ell_\gamma}} y^{-1}\, dy = \ell_\gamma$, giving

$$\int_{F_\tau} p_{\mathbb{H}^3}(t, w, \tau^m w)\, d\,\mathrm{vol}_{\mathbb{H}^3}(w) = \frac{2\pi\, e^{m\ell_\gamma}\, \ell_\gamma}{|e^L - 1|^2} \cdot \frac{2t\, e^{-t}}{(4\pi t)^{3/2}}\, e^{-(m\ell_\gamma)^2/4t}. \tag{88}$$

Using $|e^{a+ib} - 1|^2 = 2e^a(\cosh a - \cos b)$ and $\dfrac{2\pi \cdot 2t}{(4\pi t)^{3/2}} = \dfrac{1}{\sqrt{4\pi t}}$, this is equivalently

$$\int_{F_\tau} p_{\mathbb{H}^3}(t, w, \tau^m w)\, d\,\mathrm{vol}_{\mathbb{H}^3}(w) = \frac{\ell_\gamma}{2\big(\cosh(m\ell_\gamma) - \cos(m\theta_\gamma)\big)} \cdot \frac{e^{-t-(m\ell_\gamma)^2/4t}}{\sqrt{4\pi t}}. \tag{89}$$

Theorem 7.2 (Mass of subordinate Brownian loop measure on hyperbolic 3-manifolds)**.** *Let ϕ be one of the Bernstein functions considered in this paper. For any $\gamma \in \mathcal{P}_X$ and $m \ge 1$,*

$$\mu_X^\phi(\mathcal{C}_X(\gamma^m)) = \frac{2\pi\, e^{m\ell_\gamma}\, \ell_\gamma}{|e^L - 1|^2} \int_{(0,\infty)} \frac{2s\, e^{-s}}{(4\pi s)^{3/2}}\, e^{-(m\ell_\gamma)^2/4s}\, V_\phi(\,ds), \tag{90}$$

where $L = mL_\gamma = m\ell_\gamma + im\theta_\gamma$ and V_ϕ is the weighted potential measure of Definition 2.9.

Proof. Starting from (86), the inner spatial integral is taken against the Brownian kernel $p_{\mathbb{H}^3}(s, \cdot, \cdot)$ at the subordination time s, and is evaluated by (88). Substituting,

$$\mu_X^\phi(\mathcal{C}_X(\gamma^m)) = \frac{2\pi\, e^{m\ell_\gamma}\, \ell_\gamma}{|e^L - 1|^2} \int_0^\infty \frac{dt}{t} \int_{[0,\infty)} \frac{2s\, e^{-s}}{(4\pi s)^{3/2}}\, e^{-(m\ell_\gamma)^2/4s}\, \psi_t^\phi(\,ds).$$

Applying Lemma 2.11 with $h(s) = \dfrac{2s\, e^{-s}}{(4\pi s)^{3/2}}\, e^{-(m\ell_\gamma)^2/4s}$ collapses the double integral into a single integral against V_ϕ, giving (90). □

Corollary 7.3 (Mass of Brownian loop measure in free homotopy classes on hyperbolic 3-manifolds)**.** *Let $X = \Gamma\backslash\mathbb{H}^3$ be a geometrically finite hyperbolic 3-manifold, $\gamma \in \mathcal{P}_X$ a primitive closed geodesic with winding number $m \ge 1$. The mass of Brownian loop measure in a free homotopy class $\mathcal{C}_X(\gamma^m)$ is*

$$\mu_X(\mathcal{C}_X(\gamma^m)) = \frac{1}{m} \cdot \frac{1}{|e^{mL_\gamma} - 1|^2}, \qquad mL_\gamma = m\ell_\gamma + im\theta_\gamma. \tag{91}$$

Two equivalent forms are

$$\mu_X(\mathcal{C}_X(\gamma^m)) = \frac{e^{-m\ell_\gamma}}{2m\big(\cosh(m\ell_\gamma) - \cos(m\theta_\gamma)\big)} = \frac{1}{m\Big[\big(e^{m\ell_\gamma}-1\big)^2 + 4e^{m\ell_\gamma}\sin^2\!\big(\frac{m\theta_\gamma}{2}\big)\Big]}. \tag{92}$$

When $\theta_\gamma = 0$ *the holonomy term drops and the denominator becomes* $\big(e^{m\ell_\gamma}-1\big)^2$.

Proof. For pure Brownian motion, $V_\phi(\,ds) = \,ds/s$ by Example 2.10, and (90) becomes

$$\mu_X(\mathcal{C}_X(\gamma^m)) = \frac{2\pi\, e^{m\ell_\gamma}\,\ell_\gamma}{\left|e^L - 1\right|^2}\cdot\frac{2}{(4\pi)^{3/2}}\int_0^\infty s^{-3/2}\,e^{-s-(m\ell_\gamma)^2/4s}\,ds.$$

The integral $\int_0^\infty s^{-3/2}e^{-as-b/s}\,ds = \sqrt{\pi/b}\,e^{-2\sqrt{ab}}$, with $a = 1$ and $b = (m\ell_\gamma)^2/4$, equals $\sqrt{\pi}\,\frac{2}{m\ell_\gamma}\,e^{-m\ell_\gamma}$, and after cancellation gives (91). The equivalent forms (92) follow from $\left|e^{mL_\gamma}-1\right|^2 = 2e^{m\ell_\gamma}\big(\cosh(m\ell_\gamma)-\cos(m\theta_\gamma)\big)$ and, using $1-\cos(m\theta_\gamma) = 2\sin^2(m\theta_\gamma/2)$, from the expansion $\left|e^{mL_\gamma}-1\right|^2 = \big(e^{m\ell_\gamma}-1\big)^2 + 4e^{m\ell_\gamma}\sin^2\!\big(\frac{m\theta_\gamma}{2}\big)$. □

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